

On the Underlying Long Vowels in Contemporary Standard European Portuguese

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Abstract:

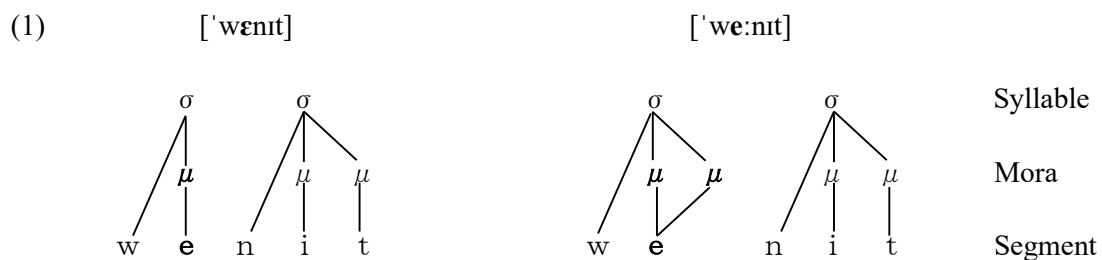
Traditionally, Portuguese linguistics has acknowledged phonetic differences in vowel length conditioned by stress in Contemporary Standard European Portuguese (EP), yet has rarely discussed the possibility that underlying vowel quantity oppositions — not directly inherited from those attested in Classical Latin, but established through phonological developments in the 14th-15th centuries — may persist in the contemporary language. In contrast, Makino (2006¹, 2006², 2007²) argued for the persistence of such underlying oppositions in vowel quantity even in Contemporary EP, seeking to account for a range of phonological phenomena: two different types of problematic alternations [é: á: ó:] ≈ [i e u] vs. [é: á: ó:] ≈ [ɛ a ɔ]; the characteristic vowel alternations in open syllables preceding (pre)palatal consonants [ʃ ʒ ʎ ɲ]; the phonetic outcomes of vowel coalescence at the phrasal level as in *fic[a]qui*; non-reduction of radical vowels in derived nouns and adjectives formed through suffixation to verbal radicals without intervening thematic vowels; and the absence of vowel reduction in unstressed syllables of derived words with negative connotations. This theoretical study revisits these issues and offers a novel account of two phenomena: the absence of vowel quality reduction in nasal vowels in unstressed syllables, and the distinctive opposition [é:] ≈ [á:] in the verb endings *-amos* and *-ámos* in the first-person plural of the present and preterite indicatives. The analysis is mainly developed within a framework of mora theory and Optimality Theory with phonological stratification, and employs a revised view of feature geometry in which the traditional [±ATR] feature is reinterpreted to represent the terminal directions of the three-stage vectorial movement leading to each height in the seven-vowel system. Rather than introducing new phonetic data — except for some instrumental measurements that appear to provide acoustic evidence for the underlying vocalic quantity opposition — this study attempts a reinterpretation of (well-known) phonetic patterns whose interrelations may not have been previously considered.

1. Introduction

In Contemporary Standard European Portuguese (henceforth EP), phonetic alternations in vowel length conditioned by stress can be observed — for example, [é:]*ma* "he/she loves; you love" versus [ɐ]*mamos* "we love" from the verb *amar* "to love". However, as far as we are aware, little to no attention has been paid to the possibility that Contemporary EP may exhibit a phonological — that is, underlying — opposition in vowel quantity. When one considers not only the stress-conditioned alternations in vowel quantity, but also the presence or absence of vowel quality alternations under the same conditioning, along with the etymological backgrounds of the relevant forms, it seems reasonable to hypothesise that a phonological opposition between short and long vowels — established by the year 1500 — may still persist at the underlying level in contemporary EP. Moreover, positing such an opposition makes it possible to offer a more unified analysis of phonetic patterns that might otherwise appear unrelated. Accordingly, this study does not seek to introduce new phonetic data, but rather constitutes a theoretical reinterpretation of well-known facts from an entirely different perspective — employing three fundamental conceptual devices: **the concept of the *mora*, the postulation of the underlying opposition between short and long vowels, and the influence of phonetic implementation of stress on the surface realisation of those underlying vowels.**

2. Phonological vowel length

In this section, we adopt the perspective that distinguishes between the segmental tier — which encodes phonemic quality — and the moraic tier — which encodes phonological quantity — in representing vowel length. Classical Latin provides a canonical case of an underlying vowel quantity opposition, as in the minimal pair *vĕnit* ['vɛnit] "he/she comes" vs. *vēnit* ['ve:nit] "he/she came". Their respective phonological representations may be schematised as in (1). (In the representations below, the final consonant /t/ is hypothetically treated as moraic; however, since the present analysis is concerned primarily with vowel quantity, the moraicity of /t/ is not crucial to the argument.)



The two forms are identical in that the segmental tier consists of the same sequence of segments — /w/, /e/, /n/, /i/, /t/ — in that order. The difference lies in whether the vocalic segment /e/ is linked to a single mora (μ) or to two moras. A mora is a unit of phonological weight and serves as a constituent of the syllable (σ) determining syllable weight — light vs. heavy. From the viewpoint of segment-to-mora association, a vowel segment (V) linked to a single μ yields a short vowel /V̥/ — e.g. /wēnit/ [ˈwēnit] — whereas one linked to two moras corresponds to a long vowel /V̄/ — e.g. /wēnit/ [ˈweːnit]. As shown in (1), onset consonants are not associated with any mora and therefore do not contribute to phonological weight.

3. Underlying moraic specification of vowel segments in Contemporary EP

In this section, we examine how vowel segments in Contemporary EP are associated with moras, from the theoretical perspective outlined in Section 2. A key piece of evidence for this investigation is the pattern of phonological alternation in 3.1.

3.1. Vowel alternations conditioned by stress and their historical background

In Contemporary EP, vowels typically exhibit phonetic alternations in both **quality** and **quantity**, as illustrated in (2), depending on whether they occur in stressed or unstressed syllables — for simplicity, the discussion in this section is restricted to open ones not preceding nasal consonants [m n ɲ] or prepalatal and palatal ones [ʃ ʒ ʎ ɲ] — within the same morpheme. Teyssier (1984: 20) refers specifically to the alternation in vowel quality as "réduction". In certain morphemes, by contrast, only alternations in **quantity** are observed, without accompanying changes in quality, as shown in (3)¹, where "≈" indicates alternation, "m.pt." stands for Middle Portuguese, "<" denotes a sound change and "□" represents a borrowing:

(2)	[é:] ≈ [i]	pr[é:]ga ≈ pr[i]gar (< m.pt. <i>pregar</i> < lat. <i>plicāre</i>) from <i>pregar</i> "to nail"
	[á:] ≈ [e]	[á:]ta ≈ [e]tar (< m.pt. <i>atar</i> < lat. <i>aptāre</i>) from <i>atar</i> "to tie"
	[ɔ:] ≈ [u]	m[ɔ:]ra ≈ m[u]rar (< m.pt. <i>morar</i> < lat. <i>morārī</i>) from <i>morar</i> "to live"

¹ In Makino (2006³), the author demonstrated acoustically that there is a statistically significant durational difference between stressed and unstressed syllables: the former are realized, on average, with approximately twice the duration of the latter. He concluded that, since no substantial durational difference was observed in the consonantal segments, this difference in syllable duration is attributable primarily to vowel length.

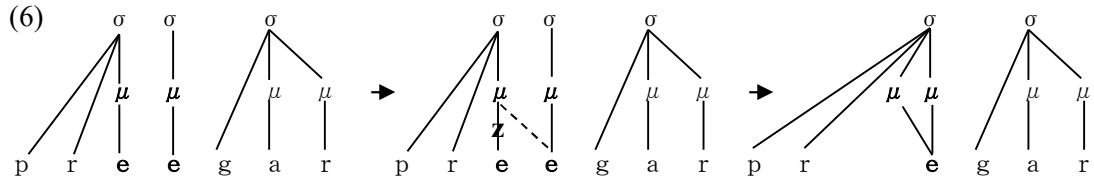
(3)	[ɛ:] ≈ [ɛ]	<i>pr</i> [ɛ:] <i>ga</i> ≈ <i>pr</i> [ɛ] <i>gar</i> (< m.pt. <i>preegar</i> < lat. <i>praedicāre</i>) from <i>pregar</i> "to preach" <i>refl</i> [ɛ:] <i>te</i> ≈ <i>refl</i> [ɛ] <i>tir</i> (< m.pt. <i>reflectir</i> □ lat. <i>reflectere</i>) from <i>refletir</i> "to reflect"
	[á:] ≈ [a]	<i>g</i> [á:] <i>nha</i> ≈ <i>g</i> [a] <i>nhar</i> (< m.pt. <i>gaanhar</i> < lat. <i>*guadaniāre</i>) from <i>ganhar</i> "to gain" [á:] <i>to</i> ≈ [a] <i>tor</i> (< m.pt. <i>acto/actor</i> □ lat. <i>āctus/āctor</i>) <i>ato</i> "act" and <i>ator</i> "actor"
	[ɔ:] ≈ [ɔ]	<i>c</i> [ɔ:] <i>ra</i> ≈ <i>c</i> [ɔ] <i>rar</i> (< m.pt. <i>coorar</i> < lat. <i>colōrāre</i>) from <i>corar</i> "to brush" <i>ad</i> [ɔ:] <i>ta</i> ≈ <i>ad</i> [ɔ] <i>tar</i> (< m.pt. <i>adoptar</i> □ lat. <i>adoptāre</i>) from <i>adotar</i> "to adopt"

From an etymological perspective, the vowel involved in the alternation in (2) descends from a monophthong (e.g. *prega* ≈ *pregar*), whereas the vowel in (3) originates in many cases from one of the following sources (4) and (5):

(4) **a sequence of two monophthongs** — resulting in a long vowel via contraction (e.g. *preega* ≈ *preegar*);

(5) **a monophthong followed by a coda consonant** — yielding a long vowel through compensatory lengthening triggered by coda consonant deletion (e.g. *reflecte* ≈ *reflectir*).

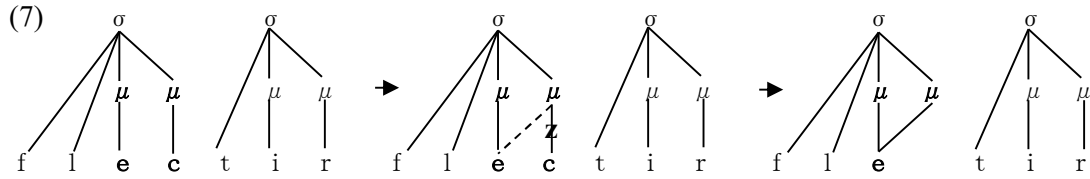
Assuming the two-tiered representation proposed in Section 2 — which distinguishes between the segmental and moraic tiers — the historical changes that the m.pt. *preegar* is likely to have undergone by "por volta de 1500" (Teyssier 1980: 43) may be schematically represented as in (6):



In the initial stage of (6), the two adjacent pretonic vowels *e* were each linked to a single mora, forming two distinct syllables: *pre*- and *-e*-. However, as a result of "contração" — a vowel coalescence process that occurred during the 14th to 15th centuries (cf. Teyssier 1980: 40-43), likely triggered by the Obligatory Contour Principle (OCP) — one of the two *es* was delinked from its associated mora, which is represented by *z* on the association line. (It is not certain which of them was actually deleted; however, given that the second *e* was the radical vowel of the verb and stressable in rhizotonic forms, it might have had more stability than the first *e*.) That mora was then relinked to the other *e*, resulting in compensatory lengthening in the second stage. Simultaneously, resyllabification took place in the third stage, yielding a new syllable *pre*-, with an underlying long vowel /ε/ as its nucleus.

In the case of *reflectir* in (3), a borrowing from Classical Latin, the historical development

follows essentially the same pattern, as illustrated in (7), where, for reasons of space, the prefix *re-* is omitted in the representation:



In this case, it is assumed that the coda consonant *-c* was deleted due to the tendency of PE to avoid closed syllables (cf. Makino 2004²: 207), which is indicated by the delinking of *-c*, and that the mora previously linked to *-c* was reassociated to the preceding $\overset{\mu}{e}$, resulting in a underlyingly long vowel $\overset{\mu\mu}{e}/$ linked to two moras.

As Teyssier (1980: 42) notes: "No século XV, quando as contracções das vogais em hiato se completaram, essas vogais deviam ser *longas e abertas*, em oposição às pretónicas simples [e] [ä] e [o], que eram *breves e fechadas*", which would be in IPA [e] [ɐ] and [o], respectively. Although the low-mid long vowel $\overset{\mu\mu}{e}/$ subsequently lost its phonetic length, it has retained its vowel quality to this day, as seen in *pr[ɛ]gar* and *refl[ɛ]tir*. On the other hand, the pretonic $\overset{\mu}{e}$ in *pregar* in (2), which was etymologically a monophthong, is assumed to have remained an underlying short vowel $\overset{\mu}{e}/$ linked to only one mora. After passing through the stage of [e] "*breve e fechada*" in the 15th century, it subsequently underwent a phonetic change "no século XVIII, provavelmente depois de 1750" (Teyssier 1980: 62), and is today realised in the form *pr[i]gar*.

This situation is assumed to hold not only for *e* / $\overset{\mu}{e}/$, but also for *a* / $\overset{\mu}{a}/$ and *o* / $\overset{\mu}{o}/$, and likewise for stressed vowels. Accordingly, between the alternations in vowel quality and quantity conditioned by the stress assignment in the Contemporary EP and their earlier states, a relationship of the type illustrated in (8) can be observed:

(8)

Middle EP (ca. 1500)		Contemporary EP
Underlying short vowels: $\overset{\mu}{\varepsilon} \overset{\mu}{a} \overset{\mu}{o}/$	$\Rightarrow$	$[\acute{e}: \acute{a}: \acute{o}:] \approx [i \text{ } \text{ɐ} \text{ } u]$
Underlying long vowels: $\overset{\mu\mu}{\varepsilon} \overset{\mu\mu}{a} \overset{\mu\mu}{o}/$		$[\acute{e}: \acute{a}: \acute{o}:] \approx [\varepsilon \text{ } a \text{ } o]$

3.2. The opposition between short and long vowels at the underlying level and their phonetic realisation

What becomes clear from the vowel alternations in Contemporary EP illustrated in (8) is the

following:

(9) The alternations in quantity — [é: á: ó:] ≈ [i ɐ u] and [é: á: ó:] ≈ [ɛ a ɔ] — are determined by the stress assignment: vowels are realised as bimoraic long vowels in stressed syllables — particularly in stressed open syllables — and as monomoraic short vowels in unstressed syllables.

(10) The alternation in quality — [é: á: ó:] ≈ [i ɐ u] — is likewise conditioned by stress, with the vowel quality *réduction* occurring in unstressed syllables.

(11) The only pattern that is synchronically not predictable is whether [é: á: ó:] in stressed syllables alternate with [i ɐ u] as in (2) or with [ɛ a ɔ] as in (3) in unstressed syllables.

The alternations of quantity in (9) and the fact that vowel quality reduction occurs in unstressed syllables, as shown in (10), are both predictable from stress assignment, and therefore pertain to the grammar. On the other hand, the pattern in (11) is unpredictable and thus constitutes lexical information. While previous studies in the generative tradition seem not to have made this point explicit, it might be inferred that both the vowels showing qualitative alternation — [é: á: ó:] ≈ [i ɐ u] — and those that do not — [é: á: ó:] ≈ [ɛ a ɔ] — are interpreted as the same underlying vowel segments /ɛ a ɔ/, with the absence of reduction in vowel quality being considered a lexical idiosyncrasy specific to the lexical item in question.

The present study, in contrast, based on the fundamental phonological principle that "**An underlying or phonological representation will contain all and only the unpredictable (distinctive) information** for each lexical item" (Kenstowicz 1994: 60), proposes the hypothesis that **the underlying opposition between short and long vowels, established by 1500, has persisted into Contemporary EP.**

The reasoning behind this hypothesis is as follows. The alternations [á:] ≈ [ɐ] and [á:] ≈ [a] are asymmetrical in that, although it is not predictable at all whether [á:] alternates with [ɐ] or [a] in unstressed syllables, the converse alternation — namely, that both [ɐ] and [a] alternate with [á:] in stressed syllables — is entirely predictable. Accordingly, if one sets aside preconceptions, it is the opposition between [ɐ] and [a] in unstressed syllables that should be considered underlying and, consequently, the changes [ɐ] → [á:] and [a] → [á:] indicate that **the underlying opposition between [ɐ] and [a] in unstressed syllables is neutralised in stressed syllables.** In other words, starting with the two unpredictable ones — [ɐ] and [a] — and aiming for the predictable one — [á:] — represents a neutralisation of the underlying opposition between the two. In contrast, the converse movement would inevitably yield a number of lexical exceptions.

Nevertheless, since [ɐ] occurs only in unstressed syllables, which suggests that it is an unstressed allophone with reduced quality, derived from some vocalic phoneme with full quality, it is implausible to assume that [ɐ] is stored in the lexicon in that form — namely, as /ɐ/. Therefore, this study assumes a full vowel /a/ — whose unmodified quality is assumed to be preserved in stressed [á:] — as the underlying representation of [ɐ]. Under this interpretation, [á:] in stressed syllables is derived by lengthening the underlying /a/ giving it one additional mora as a phonetic implementation of stress. This relationship can be summarised as in (12):

$$(12) /a^{\mu}/ \rightarrow [á:] \approx [ɐ]$$

On the other hand, if [ɐ] is assumed to be an unstressed allophone of /a^μ/, unstressed [a] in the alternation [á:] ≈ [a] cannot be interpreted as a realisation of /a^μ/, since, as argued above, [a] and [ɐ] constitute an underlying opposition. Given the alternation /a^μ/ → [á:] ≈ [ɐ] in (12), it is evident that [ɐ] in unstressed syllables is realised with a loss of the distinct quality characterising [á:] in stressed syllables. If the alternation [á:] ≈ [a] runs in parallel with this, then what [a] in unstressed syllables is losing, compared to stressed [á:], is not its quality but its quantity — more precisely, one mora of length. It follows, then, that the underlying representation of [a] is /a^{μμ}/. From this, the relationship (13) may be established:

$$(13) /a^{\mu\mu}/ \rightarrow [á:] \approx [a]$$

The same line of reasoning can be applied to the other alternations as well. Therefore, **the alternation that is both qualitative and quantitative — [é: á: ó:] ≈ [i ɐ u] — which exhibits vowel quality reduction in unstressed syllables, corresponds to the phonetic realisation of the underlying short vowels /ε^μ a^μ ɔ^μ/, whereas the uniquely quantitative alternation [é: á: ó:] ≈ [ε a ɔ], absent of such quality reduction, corresponds to the phonetic realisation of the underlying long vowels /ε^{μμ} a^{μμ} ɔ^{μμ}/.**

Under this phonological interpretation of the relevant phonetic data in (8), **both underlying short vowels and underlying long vowels undergo some form of reduction in unstressed syllables. Specifically, the underlying long vowels, which are bimoraic, weaken by losing one mora — that is, by reducing their quantity — while the underlying short vowels, which are monomoraic and thus have no mora to lose, weaken by altering their feature values — that is, by reducing their quality.**

Moreover, the presence or absence of vowel quality reduction in (10) also becomes theoretically predictable as a reflex of the opposition between underlying short vowels and underlying long vowels. That is, in unstressed syllables, the short vowels /ε^μ a^μ ɔ^μ/ — linked to a single mora in the

underlying representation — undergo vowel quality reduction and are phonetically realised as [i̯ e̯ u̯], whereas the long vowels /^{μμ}ε̯ a̯ ɔ̯/ — linked to two moras in the underlying representation — lose merely one of their moras, but retain their quality and are phonetically realised as [ε̯ a̯ ɔ̯]. Therefore, **the only information that speakers must store at the lexical level is theoretically whether or not a given lexical item contains an underlying long vowel.**

To summarise the above discussion: the phonological opposition in vowel quantity that once existed — as shown on the left-hand side of (8) — is still preserved in the underlying representation today, as an unpredictable and oppositive property of vowels; in contrast, the vowel alternations shown on the right-hand side of (8), both in quantity and in quality, are predictable from the presence or absence of stress.

A key clue in support of this hypothesis lies in the occasional occurrence of longer phonetic realisations of the vowels involved in the alternation [é̯: á̯: ó̯:] ≈ [ε̯ a̯ ɔ̯], even in phonologically unstressed syllables. For instance, the root vowel in *p[á̯:]da* (< lat. **pānāta*-) "small bread made with poor-quality flour" and its stress-shift counterpart *p[a]deiro* (< m.pt. *paadeiro* < Lat. **pānātāriu-*) "(male) baker" and its derived form *p[a]deira* "(female) baker" is often realised as long low [a̯:] in *p[a̯:]deiro* and *p[a̯:]deira* — in contrast to the consistently short mid [e̯] in *m[e̯]deira* (< lat. *māteriam*) "wood (material)"². In our personal observation, the non-reduced vowel [a̯] occurring in unstressed syllables that function as the head of a metrical foot — as in *p[a]da-* of *p[a]daria* "bakery" ← *p[á̯:]da* + *-aria* (← indicating derivation) — is often realised with a noticeably longer duration, as in *p[a̯:]daria*. This contrasts with cases like the anthroponym *C[e̯]tarina* "a female given name, equivalent to English *Catherine*", where the head of the foot *C[e̯]ta-* contains the reduced vowel [e̯], which is clearly shorter.

The following are spectrographic analysis images of the audio files available for hearing on *Infopédia Dicionários Porto Editora*, referenced in footnote 2. Figure (14) corresponds to *padaria*, and Figure (15) to *Catarina*. The analysis was conducted using Praat (version 6.4.35). Although no information is provided regarding the speaker, it is a native female speaker, and the

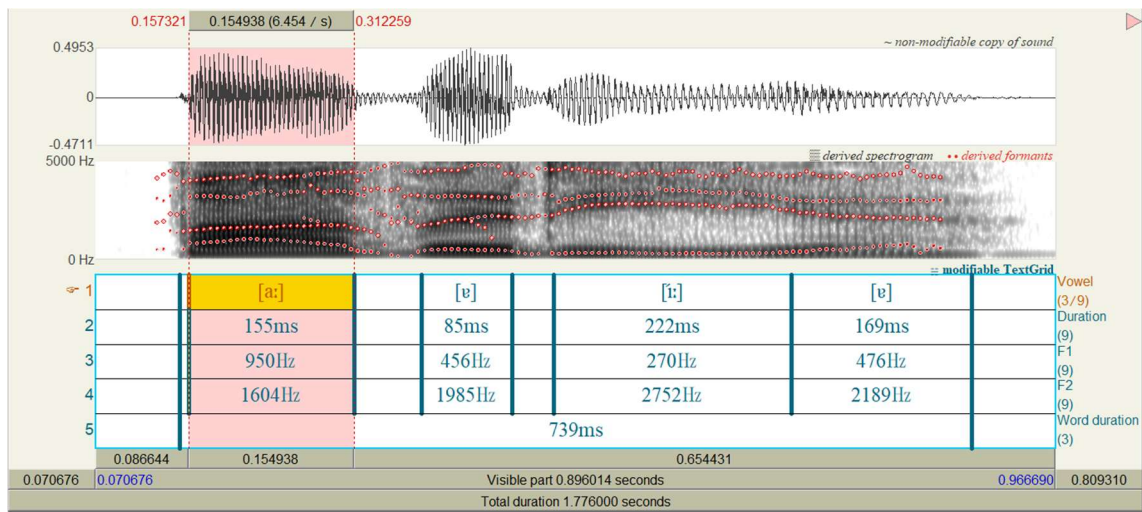
² As of 5 June 2025, the pronunciations of *pada*, *padeiro*, *padeira*, *madeira*, *padaria*, *Catarina*, *pegada*, and *pegado* can be accessed for hearing via the *Infopédia* online dictionary at the following entries:

<https://www.infopedia.pt/dicionarios/lingua-portuguesa/pada>
<https://www.infopedia.pt/dicionarios/lingua-portuguesa/padeiro>
<https://www.infopedia.pt/dicionarios/lingua-portuguesa/padeira>
<https://www.infopedia.pt/dicionarios/lingua-portuguesa/madeira>
<https://www.infopedia.pt/dicionarios/lingua-portuguesa/padaria>
<https://www.infopedia.pt/dicionarios/lingua-portuguesa/catarina>
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<https://www.infopedia.pt/dicionarios/lingua-portuguesa/pegado>

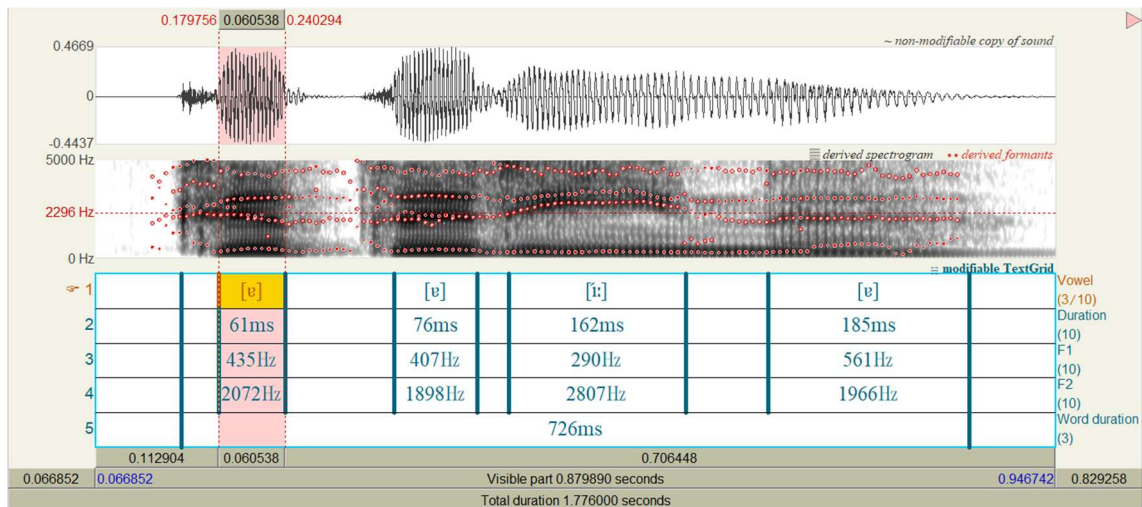
The vowel [a̯:] in *p[a̯:]daria* is particularly long and warrants close auditory attention.

same person for both recordings. The overall word durations are 739ms for *p[a:]daria* and 726ms for *C[ɐ]tarina*, and are thus nearly identical. Both words also share the same metrical structure — namely, (*p[a:].da*)('ri.a) and (*C[ɐ].ta*)('ri.na), where () indicates a metrical foot, . a syllable boundary, ' and , the primary and secondary stresses, respectively — and the two vowels under comparison both constitute the head of the secondary-stressed foot, not the primary-stressed position. In addition, the vowels that follow them are entirely identical: namely, [-ɐ-], [-í:-] and [-ɐ]:

(14) *padaria* [pa:ðɐ'ri:ɐ]



(15) *Catarina* [kɐtɐ'ri:nɐ]



The acoustic measurements of the foot *pada-* in the noun *padaria* and of the foot *Cata-* in the noun *Catarina* are given in (16) and (17), respectively:

(16)

Syllable	Vowel	Duration (ms)	F1 (Hz)	F2 (Hz)
<i>pa-</i>	[a:]	155	950	1604
<i>-da-</i>	[ɐ]	85	456	1985

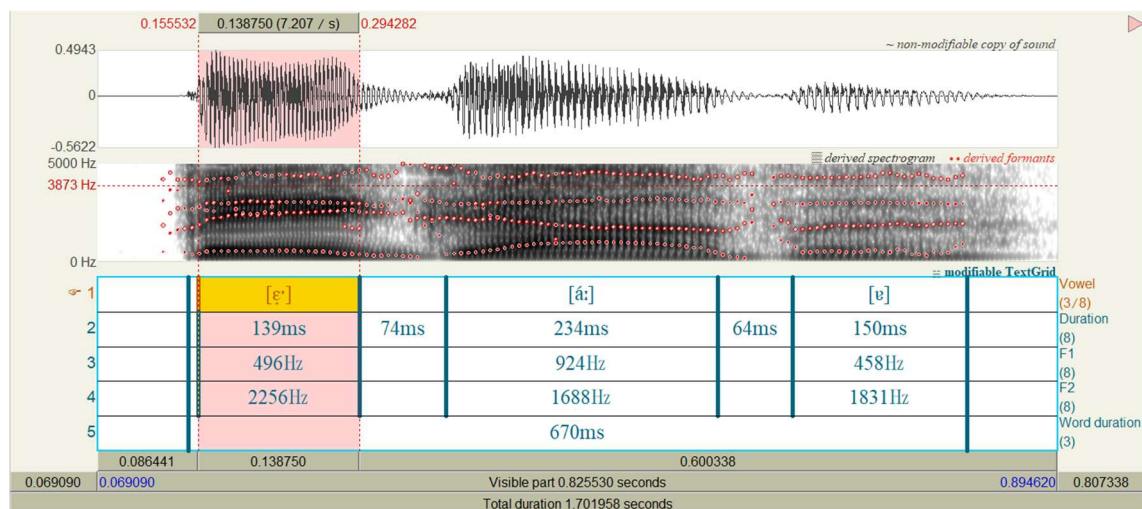
(17)

Syllable	Vowel	Duration (ms)	F1 (Hz)	F2 (Hz)
<i>Ca-</i>	[ɐ]	61	435	2072
<i>-ta-</i>	[ɐ]	76	407	1898

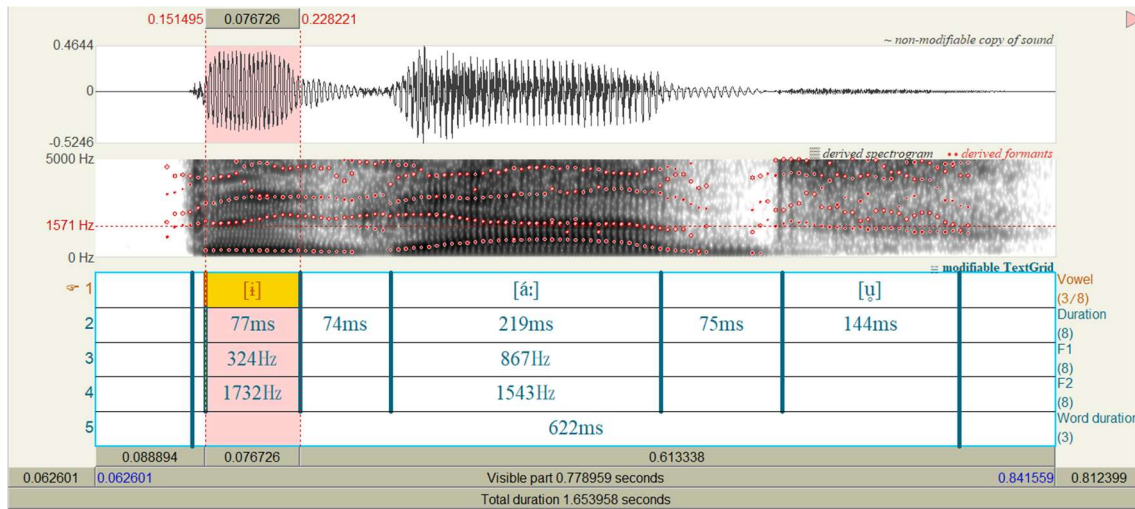
In (16), the vowel [a:] in *p[a:]daria* exhibits markedly longer duration (155ms) and a lower quality (F1 = 950Hz) than the corresponding vowel [ɐ] in *C[ɐ]tarina* (61ms, F1 = 435Hz) in (17). This striking contrast between *p[a:]daria* and *C[ɐ]tarina* is likely to provide strongly suggestive phonetic evidence that underlying bimoraic vowels /a/ are preserved not only in quality but also in duration in non-stressed foot-head positions in Contemporary EP. This would suggest that the analysis is not merely hypothetical.

Next, (18) - (20) present acoustic data from the noun *pegada* (< m.pt. *peegada* < lat. **pedicāta*-) "footprint" — derived from *pé* (< m.pt. *pee* < lat. *pede*-) "foot" — which is assumed to have /ɛ/,^μ and *pegado*, from the verb *pegar* (< lat. *picāre*) "to hold", which is assumed to contain /ɛ/:^μ

(18) *pegada* [pɛ' ga.ðɐ]



(19) *pegado* [piˈgaːd̥w]



(20)

Syllable	Vowel	Duration (ms)	F1 (Hz)	F2 (Hz)
<i>pe-</i>	[ɛː]	139	496	2256

(21)

Syllable	Vowel	Duration (ms)	F1 (Hz)	F2 (Hz)
<i>pe-</i>	[i]	77	324	1732

The analysis was conducted in the same manner as (14) and (15). As far as we can tell, it is the same native female speaker. Both words exhibit the same metrical structure — namely, (*p*[ɛː]) (*ˈga.da*) and (*p*[i])(*ˈga.do*) — with the vowels under comparison both appearing to form the head of the word-initial degenerate foot rather than occupying the primary-stressed position. Moreover, the vowels that follow them are almost identical: namely, [-á:-] and [-ɐ] or [-u]. In (18), the pretonic vowel [ɛː] in *p*[ɛː]*gada* — assumed here to be underlyingly bimoraic — exhibits nearly double the duration (148ms) and a mid front quality (F1/F2 = 496/2256Hz), compared to the unstressed vowel [i] in *p*[i]*gado* in (19), which is assumed to be underlyingly monomoraic and displays a high central quality (85ms, F1/F2 = 324/1732Hz). These vowel quantity behaviours suggest that the vowels exhibiting the alternation [é: á: ó:] ≈ [ɛ a ɔ] are underlyingly long vowels /^{μμ}ε a ^{μμ}ɔ/, which have a strong tendency to be phonetically realised as short vowels due to mora loss in unstressed positions.

Another suggestive observation is that, in Contemporary EP, length serves as the phonetic implementation of stress (cf. Martins 1983). Accordingly, it is not surprising that **underlyingly short vowels** /^μV/ **tend to be phonetically lengthened when stressed**, while **underlyingly long**

vowels $\overset{\mu\mu}{/V/}$ tend to be phonetically shortened when unstressed.

Finally, phonetic realisations of vowel coalescence at the phrasal level observed in Contemporary EP also may be taken as evidence supporting this interpretation. For example, the final vowel *-a* in *fica* — the second person singular imperative form of the verb *ficar* "to stay" — and the initial vowel *a-* in the locative adverb *aqui* "here" are both unstressed, and when pronounced in isolation, each of these is typically realised as the reduced vowel [ɐ], as in *fic*[ɐ] and [ɐ]*qui*. When the two words are combined to form the imperative sentence *Fica aqui* "Stay here", however, it is far more natural in ordinary speech to pronounce them not as *Fic*[ɐ ɐ]*qui*, but rather as *Fic*[a]*qui*, with the non-reduced low vowel [a]. This synchronic vowel crasis can be accounted for by the diagram in (22), assuming the hypothesis proposed above — namely, that an opposition has persisted between the underlying short vowels $\overset{\mu}{/ɛ} \overset{\mu}{/a} \overset{\mu}{/ɔ}/$ and the underlying long vowels $\overset{\mu\mu}{/ɛ} \overset{\mu\mu}{/a} \overset{\mu\mu}{/ɔ}/$, which, in unstressed syllables, undergo the reduction $\overset{\mu}{/ɛ} \overset{\mu}{/a} \overset{\mu}{/ɔ}/ \rightarrow [\text{i } \text{ɐ } \text{u}]$ and $\overset{\mu\mu}{/ɛ} \overset{\mu\mu}{/a} \overset{\mu\mu}{/ɔ}/ \rightarrow [\text{ɛ } \text{a } \text{ɔ}]$:

$$(22) \overset{\mu}{Fica} \overset{\mu}{a} \overset{\mu}{qui} \rightarrow \overset{\mu}{Fica} \overset{\mu}{a} \overset{\mu}{qui} \rightarrow \overset{\mu\mu}{Fic} \overset{\mu\mu}{a} \overset{\mu\mu}{qui} \rightarrow \overset{\mu\mu}{Fic} [\text{a}] \overset{\mu\mu}{qui}$$

In more explicit terms, the final *a* of *fic* $\overset{\mu}{a}$ and the initial *a* of $\overset{\mu}{a}$ *qui* are each underlying monomoraic vowels $\overset{\mu}{/a/}$. Therefore, when these words are pronounced in isolation, both vowels appear in unstressed position and undergo vowel quality reduction, being realised phonetically as [ɐ]. However, when the two words occur in sequence within a syntactic constituent, the adjacency of identical phonological elements is likely disfavoured by the OCP, leading to the deletion of the vowel quality of *a* in *fic* $\overset{\mu}{a}$, while its associated mora is retained and relinked to the following $\overset{\mu}{a}$ - of *aqui*. As a result, a *derived* long vowel $\overset{\mu\mu}{a}$ emerges in *fic* $\overset{\mu\mu}{a}$ *qui*, which is phonetically realised as a non-reduced low vowel [a] like an *underlying* long vowel $\overset{\mu\mu}{/a/}$. Other examples of the same type of coalescence — e.g. *dava-o* [ˈda:vɔ] "he/she/you used to give it", *dava-a* [ˈda:va] "he/she/you used to give it", *a* [ɐ] "to" + *a* [ɐ] "the" → *à* [a] "to the", and *a* [ɐ] "to" + *aquilo* [ɐˈki:lu] "that" → *àquilo* [aˈki:lu] "to that" — will be dealt with in Section 6.3.

In sum, it is assumed that, in EP, the historical strengthening of stress — which is reflected in the fact that "O português vem sendo trabalhado desde séculos pela tendência ao enfraquecimento das vogais átonas" (Teyssier 1982: 66) — has led to a situation in which, **in stressed syllables, the lengthening of short vowels has neutralised the underlying opposition in vowel quantity, making it no longer perceptible at the phonetic level.** By contrast, **in unstressed syllables, the reduction of vowel quality in short vowels and the reduction of quantity in long vowels has caused the underlying opposition in vowel**

quantity to be phonetically implemented by means of vowel quality³.

4. From underlying to surface representations

The next issue concerns how to account for the mapping of vowel segments from their underlying representations to their surface ones: $/\overset{\mu}{\varepsilon} \overset{\mu}{a} \overset{\mu}{\omega}/ \rightarrow [\acute{\varepsilon}: \acute{a}: \acute{\omega}:] \approx [i \text{ } \text{v} \text{ } u]$ and $/\overset{\mu\mu}{\varepsilon} \overset{\mu\mu}{a} \overset{\mu\mu}{\omega}/ \rightarrow [\acute{\varepsilon}: \acute{a}: \acute{\omega}:] \approx [\varepsilon \text{ } a \text{ } \omega]$. In addition to $/\overset{\mu}{\varepsilon} \overset{\mu}{a} \overset{\mu}{\omega}/$, the underlying inventory of short vowels in Contemporary EP also includes $/\overset{\mu}{i} \overset{\mu}{e} \overset{\mu}{o} \overset{\mu}{u}/$. Taking these into account, it appears that in stressed open syllables — particularly not before [+nasal] consonants, [−back, +high] consonants nor vowels⁴ — and in pretonic open syllables — especially when not in absolute word-initial position — the following complex mapping generally occurs: $/\overset{\mu}{i} \overset{\mu}{e} \overset{\mu}{\varepsilon} \overset{\mu}{a} \overset{\mu}{\omega} \overset{\mu}{o} \overset{\mu}{u}/ \rightarrow [\acute{i}: \acute{e}: \acute{\varepsilon}: \acute{a}: \acute{\omega}: \acute{o}: \acute{u}:] \approx [i \text{ } i \text{ } \text{v} \text{ } u \text{ } u \text{ } u]$. This paper proposes that these mappings, including those described earlier, can be systematically derived within a framework of OT with phonological stratification, as shown in (23):

(23)

$/\overset{\mu}{i} \overset{\mu}{e} \overset{\mu}{\varepsilon} \overset{\mu}{a} \overset{\mu}{\omega} \overset{\mu}{o} \overset{\mu}{u}/$	$/\overset{\mu}{i} \overset{\mu}{e} \overset{\mu}{\varepsilon} \overset{\mu}{a} \overset{\mu}{\omega} \overset{\mu}{o} \overset{\mu}{u}/$	$/\overset{\mu\mu}{\varepsilon} \overset{\mu\mu}{a} \overset{\mu\mu}{\omega}/$	$/\overset{\mu\mu}{\varepsilon} \overset{\mu\mu}{a} \overset{\mu\mu}{\omega}/$	Underlying representation
í é é á ó ó ú	—	é: á: ó:	—	Stress assignment
—	i i i v u u u	—	—	Vowel quality modification
í: é: é: á: ó: ó: ú:	—	—	ε a ω	Vowel quantity adjustment
[í: é: é: á: ó: ó: ú:]	[i i i v u u u]	[é: á: ó:]	[ε a ω]	Surface representation

Since the reduction of the vowel quality of underlying short vowels occurs in unstressed syllables, syllabification and stress assignment must already have taken place prior to that reduction stage. Furthermore, since vowel quality reduction in unstressed syllables affects only underlying short vowels, the underlying distinction in vowel quantity must still be preserved at that stage. Vowel quantity adjustment, then, must take place only subsequently. The processes of

³ It may be observed that the present analysis, in adopting a moraic approach to the vowel system of EP, might in some respects parallel the later refinements seen in Japanese phonology, where exclusively mora-based analyses — initially formulated by native speakers of Japanese, for whom, whereas the mora is self-evident, the syllable does not naturally and intuitively emerge as both a phonetic and phonological unit; indeed, for speakers of a language where the syllable is self-evident, it must be surprising that it is practically impossible for the average native speaker of Japanese to identify or count syllables in speech without explicit instruction — have gradually been balanced and supplemented by syllabic perspectives through external reinterpretations. The present proposal likewise seeks not to import foreign models, but rather to uncover the underlying prosodic organisation inherent to the EP system itself.

⁴ In Portuguese, the syllable-initial [+nasal] consonants are [m n ɲ]. In this paper, syllable-initial [ʃ ʒ ʎ ɲ] are considered doubly articulated consonants, specified as [−back, +high] in their secondary articulation, as shown in Section 6.2.

word formation and the accompanying deletion, epenthesis, feature change, syllabification and stress assignment will be discussed in separate papers; in this paper, it is assumed that the next successive phonological strata (24) and (25) follow after stress assignment:

(24) Vocalic Quality Modification Stratum

In this stratum, the following set of constraints is assumed to be interacting:

The inviolable constraint IDENT-QUANTITY;

IDENT(lowVhigh) >> *[+low]/V[−stressed, μ] >> *[−high]/V[−stressed, μ] >> IDENT(high);

*[+low]/V[−stressed, μ] >> IDENT(low);

*[±ATR]/V[−stressed, μ] >> IDENT(ATR);

*[−back]/V[−stressed, μ]__([−sonorant])# >> IDENT(high^back) >> *[−back]/V[−stressed, μ] >> IDENT(back);

IDENT(high), IDENT(low), IDENT(ATR), IDENT(round) >> *[+high], *[−high], *[+low], etc.

(25) Vocalic Quantity Adjustment Stratum

In this stratum, the following set of constraints is assumed to be interacting:

The inviolable constraint IDENT-QUALITY;

MAX, DEP >> *σ[μμμ] >> *'σ[μ] >> *V[μμ] >> MAX(μ);

*'σ[μ] >> *V[μμ], DEP(μ), DEP(high), DEP(ATR);

*[−low, −high, −back, μμ]_.V >> DEP(high), DEP(ATR);

*σ[C₀V₁V₂[−high, ±ATR]] >> DEP(high), DEP(ATR)

The definitions of these constraints, as well as the justification for their rankings, will be discussed in Sections 4.1 and 4.2.

4.1. Vocalic Quality Modification Stratum

In this stratum, the short vowels ^{μ μ μ μ μ μ}i ē ε a ɔ o u — hereafter simplified as i e ε a ɔ o u — generally undergo changes in quality in unstressed syllables. This study limits its discussion to the following types of unstressed syllables: non-absolute word-initial, word-medial and word-final

open syllables, as well as non-absolute word-initial, word-medial and word-final syllables closed by [ʃ ʒ], and non-absolute word-initial and word-medial syllables closed by [ɾ]⁵. By contrast, the long vowels ^{μ μ μ}ε a ɔ — henceforth ε: a: ɔ: — pass through this stratum without undergoing any changes in either quality or quantity, and will instead be subject to quantitative adjustment in the next stratum, which will be discussed in 4.2. However, before turning to the feature value changes affecting the short vowels here, it is necessary to first consider the nature of tongue-height or aperture oppositions among the vocalic segments of Contemporary EP.

In representative generative phonological studies of Portuguese — namely, Andrade (1977: 96-100), Mateus (1982: 25-26), Mateus & Andrade (2002: 31-35), and Mateus et al. (2003: 998-1001) — an analysis such as (26) has been adopted⁶.

(26)

high				low
+	/i/	[i]	/u/	—
—	/e/	[e]	/o/	
	/ɛ/	/a/	/ɔ/	+

This analysis of vowel tongue height or aperture is simple and free of redundancy, but it presents several problems in terms of phonological theory⁷. First, if /ε a ɔ/ formed a natural class on the basis of sharing the feature [−high, +low] in terms of vowel tongue height or the correlate

⁵ In unstressed syllables closed by /l/, the vowel reduction described in (23) does not occur (e.g. *feltar* "to felt" /feltr-a-r0/ → [fel'tra:r]; *selvagem* "wild" /selv-aʒen/ → [seɫ'va:ʒẽi]; *faltar* "to lack" /falt-a-r0/ → [fal'ta:r]; *voltar* "to return" /vɔlt-a-r0/ → [voɫ'ta:r]~[vɔɫ'ta:r]; *amável* "kind" /am-a-vɛl-0/ → [v'ma:vɛɫ]; *Setúbal* "a Portuguese city's name" /setubal-0/ → [si'tu:βaɫ]). In unstressed syllables closed by /n/, the vowel+/n/ sequence — which bears two moras — is assumed to undergo vocalic nasalisation and nasal deletion in an earlier stratum — e.g. /in/ → ɪ̃: — and thus enters the current stratum as a bimoraic nasal vowel, as discussed in Section 6.1. Other environments — namely, absolute word-initial unstressed open syllables as well as absolute word-initial unstressed syllables closed by /s/, /r/, or /n/ and word-final syllables closed by /r/ or /n/ — exhibit reduction patterns that vary depending on whether the syllable is open or closed, and on the nature of the coda consonant; due to the considerable complexity involved, these cases will not be discussed here.

⁶ In Mateus & Andrade (2002), /i/ is treated as the least marked vowel in Portuguese and is subject to underspecification. As a result, /i/ and /u/ are not specified for either [high] or [low] in the underlying representation, and /e/ and /o/ are specified only as [−high], while /ɛ/, /a/, and /ɔ/ are specified only as [+low].

⁷ The reason why an analysis such as that shown in (26) has been adopted likely lies in the fact that, when following Halle's articulator-based model and analysing [i u] as [+high], the set [e o ε ɔ] as [−high, −low], and [a] as [+low], there is no alternative but to define the opposition between [e ɔ] and [ε ɔ] in terms of [+ATR] versus [−ATR] opposition. For native researchers, however, the opposition between [e ɔ] and [ε ɔ] — as suggested by orthographic conventions, that is, the acute and circumflex accents (á/â, é/ê, ó/ô), the denomination of "vogais fechadas/semifechadas/semiabertas/abertas", and their own phonological analyses — is intuitively one of tongue-height or aperture, and it is likely that there has been hesitation to introduce [±ATR], in its original sense of tongue root advancement, into the analysis of their own native language.

aperture, then from the perspective of phonological theory, one would expect them to behave uniformly with respect to processes of vowel raising in unstressed syllables, since phonological features are theoretical constructs introduced precisely to account for the common behaviour of a set of segments. In pretonic CV syllables, /a/ is raised to the mid vowel [ɐ], specified as [−high, −low]. If /ɛ/ and /ɔ/ formed the same natural class with /a/, one would expect them to be raised — with the feature value change [+low] → [−low] — to the mid vowels [e] and [o], likewise specified as [−high, −low], as is [ɐ]. Nevertheless, they are actually raised to the high vowels [i] and [u] — that is, vowels specified as [+high, −low], respectively. This difference in pattern suggests that these three vowels do not belong to the same natural class in terms of tongue height or aperture.

Second, in EP, as will be addressed in Section 6.4, /a/ undergoes raising to [é:] in stressed CV syllables preceding [+nasal] consonants — namely, [m n ɲ] — whereas /ɛ/ and /ɔ/ retain their underlying tongue height or aperture and are realised as [é:] and [ó:] — e.g. *cama* "bed" /kam-a/ → ['kə:mɐ]; *remo* "oar" /rɛm-o/ → ['rɛ:mu]; *fome* "hunger" /fɔm-e/ → ['fɔ:mɪ]. Within the aforementioned model, this gives rise to a further asymmetry: although /ɛ a ɔ/ are all analysed as [+low] and thus constitute a natural class, /a/ behaves differently from /ɛ/ and /ɔ/ with respect to yet another type of raising. Consequently, if one assumes that /ɛ a ɔ/ form a natural class by virtue of their [+low] specification, one must then explain why /a/ undergoes raising while /ɛ/ and /ɔ/ do not in this context, based on some other feature that distinguishes /a/ from the other two within the [+low] class. In their descriptive framework, /ɛ/ is distinguished from /a/ only by its [−back] specification and /ɔ/ is distinguished from /a/ only by its [+round] specification. It would follow, therefore, that the features [−back] in /ɛ/ and [+round] in /ɔ/ somehow inhibit the [+low] → [−low] change in stressed CV syllables preceding [+nasal] consonants — yet such an explanation seems rather implausible.

Third, in pretonic CV syllables in Brazilian Portuguese — henceforth, BP — /a/ remains as [a], specified as [+low], whereas /ɛ/ and /ɔ/ are raised to the high-mid vowels [e] and [o], respectively, specified as [−low]. This pattern too is difficult to account for under the assumption that /ɛ a ɔ/ constitute a natural class of [+low] vowels.

Fourth, in BP, the word-final vowel sequences /-e-a/ and /-ɛ-a/ are distinguished both phonemically and phonetically, as exemplified by *teia* "cobweb" /te-a/ → ['tɛ̃j.ɐ] and *ateia* "atheist" /a-tɛ-a/ → [a'tɛ̃j.ɐ], respectively. In EP, however, this phonological opposition is neutralised at the surface level: both /-e-a/ and /-ɛ-a/ are realised as [−ɛ̃j.ɐ], as in *teia* /te-a/ → ['tɛ̃j.ɐ] and *ateia* /a-tɛ-a/ → [ɐ'tɛ̃j.ɐ]. That is, [ɐ] functions as a **mid** central vowel that

corresponds to both the **high-mid** front vowel /e/ and the **low-mid** front vowel /ɛ/ and, as such, it stands in opposition to the **low** central vowel /a/, as in *má* "bad" (fem.) /ma-a/ → [ˈma:] and *faia* "beech" /fai-a/ → [ˈfai̯.ə]. The centralisation [−back] → [+back] of /e/ and /ɛ/ before the seemingly epenthetic [i̯] may be attributed to an OCP-enforced constraint against adjacent [−back] features, since both /e/ or /ɛ/ and [i̯] are all specified as [−back]. Accordingly, if /ɛ/ were [+low] just like /a/, there would be no explanation for why *ateia* is not realised as *[aˈtai̯.ə] simply with the change [−back] → [+back], like /a-tɛ-a/ → aˈtɛi̯.ə → *[aˈtai̯.ə].

From a theoretical perspective, it seems reasonable to posit that the low vowel /a/ occupies the foundational position in the vowel system — articulatorily the lowest and acoustically the most salient vowel. The high vowels /i/ and /u/ then arise as maximally opposed poles relative to /a/, completing a triangular system that ensures maximal perceptual opposition. Subsequently, the addition of /e/ and /o/ serves to fill the **mid**-space between these extremes, producing the commonly observed five-vowel system. Only later, as a form of system refinement, do languages introduce further oppositions between **high-mid** and **low-mid** vowels — /e/ vs. /ɛ/, and /o/ vs. /ɔ/ — distinctions that are structurally fragile and often prone to neutralisation, as in French and EP, for instance. This trajectory — from /a/ to /i u a/, then to /i u e o a/, and finally to /i u e o ɛ ɔ a/ — appears to align closely with both the gradual subdivision of the articulatory-acoustic vowel space and cross-linguistic patterns of oppositional vowel relationships.

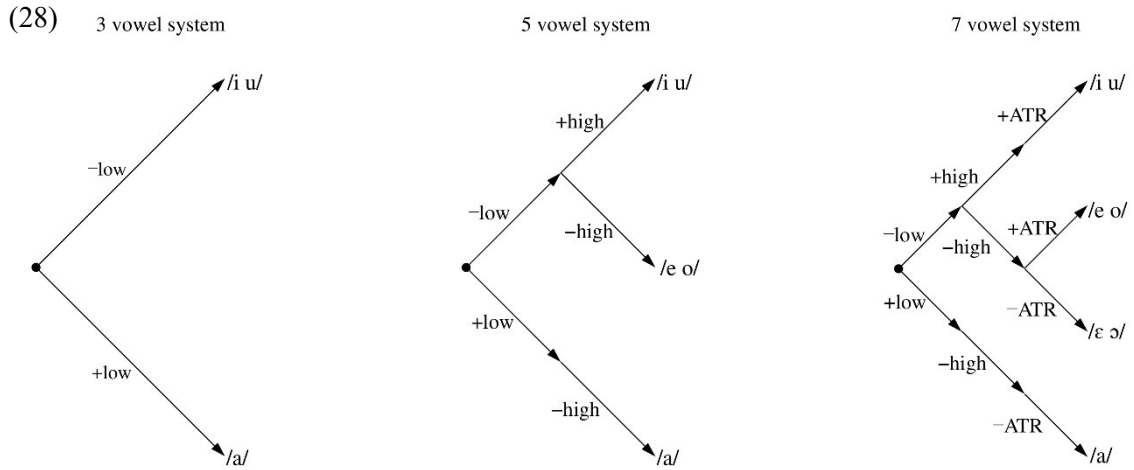
Based on the above discussion, then, it seems more appropriate to analyse the vowel tongue-height opposition among EP vocalic segments in terms of [±high] and [±low], as in (27):

(27)

high				low
+	/i/	[i̯]	/u/	—
—	/e/	[ɐ]	/o/	
	/ɛ/		/ɔ/	
		/a/		+

The problem with the system in (27) is that it lacks a feature capable of encoding the opposition — in tongue height or aperture — between /e o/ and /ɛ ɔ/, as indicated by the shaded cells. Given that the aforementioned studies consistently analyse this opposition as one between [−low] and [+low], without no reference to [±ATR], and that other researchers within the structuralist-functionalist tradition (cf. Barbosa 1983, 1994¹, 1994²) have likewise consistently treated it as an opposition in aperture, it appears that, for native-speaker researchers, this is self-evidently a tongue-height-based or aperture-based opposition.

Thus, in this paper, the feature $[\pm ATR]$ — which is assumed to be dominated by the Pharyngeal Cavity node in Halle's feature geometry — is employed *in italics> as a convenient label for representing **an additional tongue-height or aperture opposition within the $[-low, -high]$ region**. This paper proposes the following (28) systems of vowel tongue height or aperture:*



In (28), $[\pm low]$, $[\pm high]$ and $[\pm ATR]$ are reinterpreted as features that encode the conceptual oppositive movement of the tongue to define and differentiate the tongue height or aperture of vowel segments within the vowel articulatory space — starting from a neutral position, represented by $\bullet$ — toward their specific tongue-height targets within the system: high /i, u/, high-mid /e, o/, low-mid /e ɔ/ and low /a/.

First, $[-low]$ and $[+low]$ are taken to encode upward and downward movement, respectively, from the neutral position. Whereas in traditional systems $[-low]$ expresses a neural command not to lower the tongue from the neutral position and, correlatively, the tongue position itself, in the present system it represents upward movement from that position. In a three-vowel system such as /i u a/, the endpoints of the $[\pm low]$ movement coincide with the upper and lower bounds of the articulatory space, thereby determining the tongue height or aperture of each vowel: /i u/ are specified as $[-low]$, while /a/ is specified as $[+low]$ ⁸.

⁸ The vowels /a/ and /i u/ stand in a binary opposition in terms of tongue-height or aperture. Accordingly, if this opposition is to be represented by features that may assume both positive and negative values, a single feature is sufficient, and employing two features would be theoretically redundant. It should be noted that, in this model, the starting point is not taken to be $[\pm high]$, because — as will become clear in the next paragraph — doing so would require describing /e o/ as $[+high, +low]$, thereby breaking terminological continuity with the traditional model.

In a five-vowel system such as /i u e o a/, by contrast, the feature [$\pm$ low] alone does not suffice to determine the tongue height of each segment. From the endpoint of [$\pm$ low] movement, a subsequent [$\pm$ high] movement begins: [+high] encodes upward movement and [−high] encodes downward movement. Thus, /i u/ reach their respective endpoints by undergoing a continuous upward movement first encoded by [−low] and then by [+high], while /a/ reaches its endpoint through a continuous downward movement encoded by [+low] followed by [−high]. In both cases, the endpoint of the movement coincides with either the upper or lower boundary of the vowel articulatory space, thereby terminating the movement. On the other hand, the mid vowels /e o/, specified as [−low, −high], first rise under [−low] and then change direction under [−high], resulting in an endpoint located at an intermediate tongue height between that of /i u/ and /a/. Note, however, that in actual articulation, the tongue does not first rise and then fall for /e o/; rather, they are produced directly at a mid-tongue position. The sequence [−low, −high] represents not a literal articulatory trajectory of the tongue, but rather the abstract vectorial movement by which the tongue height of each vowel segment is determined. The markedness of the mid vowels /e o/ in relation to both the high vowels /i u/ and the low vowel /a/ is thus captured by the assumption that the vectorial movement leading to their endpoints includes a point of inflexion — that is, a structural complexity.

Finally, in a seven-vowel system such as /i u e o ε ɔ a/ in Portuguese, the features [$\pm$ low] and [$\pm$ high] alone are insufficient to determine the tongue height of each segment. From the endpoint of the [$\pm$ high] movement, a further movement encoded by [$\pm$ ATR] begins. [+ATR] represents an upward movement, while [−ATR] represents a downward movement. As already noted, it is important to bear in mind that in this paper, [ATR] in italics does not refer to tongue root advancement in its original sense. Accordingly, /i u/ undergo a continuous upward movement in the sequence [−low, +high, +ATR], reaching their final endpoint; /a/, on the other hand, undergoes a continuous downward movement in the sequence [+low, −high, −ATR], likewise reaching its endpoint. In both cases, the endpoint coincides with the upper and lower bounds of the vowel articulatory space, and thus the movement terminates. The vowel /e o/, specified as [−low, −high, +ATR], initially ascends by [−low], then changes direction to descend via [−high] and then ascends again via [+ATR], ultimately reaching a high-mid tongue height as the endpoint. By contrast, /ε ɔ/, specified as [−low, −high, −ATR], first ascend via [−low], then descend via [−high] and then continue to descend via [−ATR], ultimately reaching a low-mid tongue height as the endpoint. In summary:

/a/ [+low, −high, −ATR] : **neutral position** → **descent** → **descent** → **descent**

/i u/ [–low, +high, +ATR] : **neutral position** → **ascent** → **ascent** → **ascent**

/ε ɔ/ [–low, –high, –ATR] : **neutral position** → **ascent** → **descent** → **descent**

/e o/ [–low, –high, +ATR] : **neutral position** → **ascent** → **descent** → **ascent**

To summarise all of the above, each of the features [±low], [±high] and [±ATR] corresponds to a single vectorial component of the movement by which the tongue height of each vowel — and correlatively its aperture — is determined. Rather than functioning as static coordinates that locate the vowel within the vowel space, these features behave as vectors that define the directional trajectory towards the vowel's articulatory endpoint in tongue height or aperture.

What must be noted here is the asymmetry in markedness between the mid vowels /e o/ and /ε ɔ/ in the context of a seven-vowel system. While both classes of vowels are marked in that their tongue height is determined by a vectorial movement with an inflexion point, /ε ɔ/ are less marked than /e o/ in that they involve only one inflexion point, whereas /e o/ involve two. From this perspective, the vowels in a seven-vowel system that correspond to /e o/ in a five-vowel system are not /e o/ — they are only graphically identical — but rather /ε ɔ/. In Contemporary EP, when scholarly words of Latin origin contain stressed vowels written as *e* or *o*, their quality is consistently low-mid [ε] and [ɔ] and not high-mid [e] and [o], regardless of vowel quantity in the Latin source. In other words, the opposition between /e o/ and /ε ɔ/ is neutralised in scholarly words. The fact that the outputs in such cases are [ε ɔ] — [–low, –high, –ATR] — and not [e o] — [–low, –high, +ATR] — may be attributed to the relative unmarkedness of [ε ɔ] in stressed syllables, whose height-defining vectorial movements contain only a single inflexion point.

Barbosa (1983: 76) notes that "/a/ est probablement légèrement plus fermé dans *sei* «je sais» que dans *cama* «lit»", and this, too, can be accounted for within the framework of the schema in (28). The vowel /a/ — namely [ɐ], as he considers the stressed *-e-* in *sei* and the stressed *-a-* in *cama* to be phonetic realisations of the same phoneme /a/ — can be understood, in the case of *s[ɛ̃]i*, as an allophone that results from the following process (29) and (30):

(29) The feature [–back], shared by /e/, specified as [–low, –high, +ATR, –back, –round], and /ε/, specified as [–low, –high, –ATR, –back, –round], is prohibited by the OCP from occurring adjacent to the same [–back] feature of a following [–i]. As a result, /e/ and /ε/ undergo a change from [–back] to [+back].

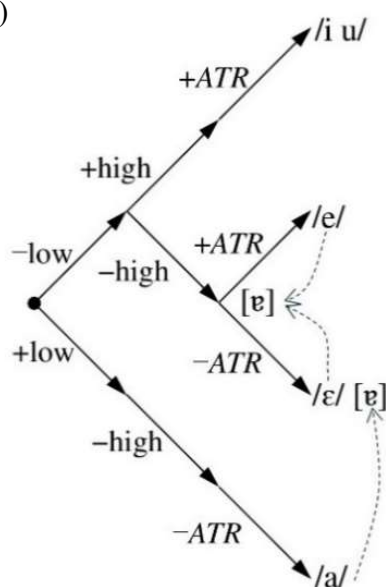
(30) Through the change [–back] → [+back], /e/ and /ε/ move into the mid central region,

defined by $[-\text{low}, -\text{high}, +\text{back}, -\text{round}]$. In this region, the distinctive opposition in $[\pm\text{ATR}]$ becomes impossible to maintain, leading to the deletion of the $[\pm\text{ATR}]$ specification from both segments. As a result, they are both underlying segments realised as $[\text{v}]$, a vowel lacking any $[\text{ATR}]$ specification and characterised by $[-\text{low}, -\text{high}, +\text{back}, -\text{round}]$.

The forms discussed above — *teia* /te-a/ → $[\text{'t}\text{ɛ}_\text{ɪ}\text{v}]$ and *ateia* /a-tɛ-a/ → $[\text{v}'\text{t}\text{ɛ}_\text{ɪ}\text{v}]$ — are precisely examples of this neutralisation of opposition.

On the other hand, $[\text{v}]$ in $c[\text{v}]:ma$, as examined in Section 6.4, is an allophone resulting from nasal resonance in /a/, triggered by the spreading of the $[\text{+nasal}]$ feature of $[\text{m}]$, which diverts part of the pulmonic airstream into the nasal cavity. This reduces the amount of airflow usable

(31)



for oral articulation and appears to make the $[\text{+low}]$ region available. Consequently, in the underlying form /a/ — specified as $[\text{+low}, -\text{high}, -\text{ATR}, +\text{back}, -\text{round}]$ — only the feature $[\text{+low}]$ is altered, changing to $[-\text{low}]$, while all other features remain unaffected. Thus, the resulting $[\text{v}]$ in this context is specified as $[-\text{low}, -\text{high}, -\text{ATR}, +\text{back}, -\text{round}]$.

It is evident from (31) that this $[\text{v}]$ and the one derived from the neutralisation of the opposition between /e/ and /ɛ/ differ in tongue height. As we shall see immediately below, the interpretation of tongue-height or aperture degree in (28) also

provides a straightforward explanation for the tongue-height change involved in the reduction of /a/ to $[\text{v}]$ in unstressed syllables.

Returning to the feature changes affecting unstressed short vowels in this stratum, they are illustrated in (32), where the altered feature values are shaded for clarity:

(32)

	high	low	ATR	back	round
i	+	—	+	—	—
e	—	—	+	—	—
ɛ	—	—	—	—	—
a	—	+	—	+	—
ɔ	—	—	—	+	+
o	—	—	+	+	+
u	+	—	+	+	+

→

	high	low	ATR	back	round
i	+	—	0	—	—
ɪ	+	—	0	+	—
ɪ	+	—	0	+	—
ɐ	—	—	0	+	—
u	+	—	0	+	+
u	+	—	0	+	+
u	+	—	0	+	+

The underlying four degrees of tongue height — /i u/ vs. /e o/ vs. /ε ɔ/ vs. /a/ — are reduced to only a two-degree opposition in unstressed position: [i ɪ u] vs. [ɐ]. Since [+low] is no longer active in the system and all the derived vowels share [−low], the resulting opposition in tongue height is effectively reduced to a two-way contrast encoded solely by [±high]. Accordingly, the specification [±high] alone suffices, and [±ATR] becomes unnecessary and is left unspecified. In fact, the unstressed [ɐ], specified as [0ATR], differs in tongue height from the stressed [é:] before heterosyllabic nasals [m n ɲ], which is specified as [−ATR]. Specifically, unstressed [ɐ] has a tongue height closer to that of IPA [ə], a higher central vowel. Emiliano (2009: 31-33) provides crucial observations on this point: while stressed [é:] aligns in height with [ε] and [ɔ], unstressed [ɐ] is slightly higher and should be transcribed as [ə], positioned between [e o] and [ε ɔ] in terms of tongue height. For example, the word *cama* should ideally be transcribed not as ['kɐ:mɐ] but as ['kɐ:mə], according to the author. He noted (2009: 31): "Não é habitual entre nós a utilização do símbolo ə para representação da vogal oral central média que ocorre em posição átona (como alofone de /a/). ... A manutenção do símbolo ɐ na transcrição da vogal átona central média neste livro resulta de simples conformidade relativamente à tradição estabelecida, apesar de o símbolo ə ser mais adequado. ... A **vogal átona central média [ɐ]** é, na realidade, ligeiramente mais fechada (e menos periférica) que a **vogal tónica central média representada pelo mesmo símbolo.**" (The bold text is as in the original.)

In this weak position, vowel reduction can be understood to proceed logically as follows:

(33) The rising of the low vowel /a/ occurs, resulting in /a/ → [ɐ], which corresponds to the feature-value change [+low] → [−low].

(34) As a consequence of (33), /e ε/, which share [−low, −high, −round] with the resulting [ɐ], undergo a feature-value change from [−high] to [+high] in order to preserve their underlying opposition in tongue height with /a/, now realised as [ɐ].

(35) Furthermore, in order to maintain their underlying opposition to /i/, which surfaces as [i], /e ε/ also shift from [−back] to [+back], ultimately mapping /e ε/ to [ɪ].

(36) The feature [±ATR], which underlyingly distinguishes between /e/ and /ε/, introduces a fine-grained tongue-height opposition only into the mid region [−low, −high]. Once both /e/ and /ε/ shift from [−high] to [+high], this opposition becomes untenable and thus neutralised, resulting in [±ATR] → [0ATR].

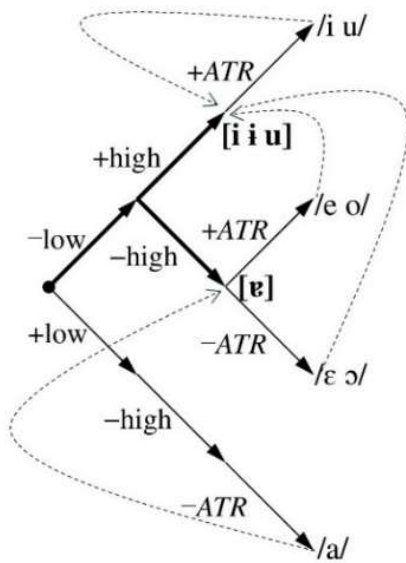
(37) The back round vowels /o ɔ/, which are specified as [−low, −high] and thus form a natural class with /e ε/ in terms of tongue height, undergo parallel changes: [−high] →

[+high] and [$\pm ATR$] $\rightarrow$ [0*ATR*], and are therefore mapped to [u].

(38) In order for /o ɔ/, now realised as [u], to preserve their opposition to the underlying /u/, also realised as [u], they would furthermore need to change [+round] to [−round]; nevertheless, doing so would yield [i], thereby neutralising the underlying opposition between /o ɔ/ and /e ε/, which are likewise realised as [i]. As a result, the system opts instead to neutralise the opposition between /o ɔ/ and /u/, all of which are realised as [u].

Employing the diagram in (28), this vowel reduction can be schematised as in (39), where the dashed curves represent the process of vowel reduction:

(39)



The conception of the vowel opposition in tongue-height degree or aperture degree, introduced in (28), provides a straightforward description of the vowel quality reduction process in unstressed syllables in (33)-(39). **In terms of tongue height or aperture, this reduction is extremely simple: it consists merely of deleting all the [*ATR*] values and raising the tongue height by one step: [+low] becomes [−low] for /a/ and [−high] becomes [+high] for /e o ε ɔ/. For /e ε/, moreover, this change is accompanied by a shift from [−back] to [+back].**

Accordingly, the vowel quality changes listed in (32) can be classified into two distinct types. The first type concerns features related to the tongue height:

(40) Raising of low vowels: [+low] $\rightarrow$ [−low] (a $\rightarrow$ ɐ)

(41) Raising of mid vowels: [−high] $\rightarrow$ [+high] (e ε o ɔ $\rightarrow$ i i u u)

(42) Non-utilisation of [*ATR*] within the system: [$\pm ATR$] $\rightarrow$ [0*ATR*] (i e ε a ɔ u $\rightarrow$ i i i ɐ u u u)

The second type pertains to the tongue backness:

(43) Centralisation of mid front vowels: [−low, −high, −back] $\rightarrow$ [+back] (e ε $\rightarrow$ i)

From the data in (40) and (41), the descriptive generalisation in (44) can be formulated, from which the relevant constraints and their ranking can be inferred. "A *descriptive generalisation*

is", as McCarthy (2008: 32) says, "the essential intermediate step between data and analysis. Good descriptive generalisations are accurate characterizations of the systematic patterns that can be observed in the data."

(44) In unstressed syllables, short low vowels specified as [+low] and short mid vowels specified as [−low, −high] are prohibited. This requirement is enforced by the unfaithful mappings [+low] → [−low] and [−high] → [+high] (e.g. *bolacha* "biscuit" /bol-aʃ-a/ → bo'la.ʃa → bu'la.ʃɐ vs. *bu'la.ʃa; *bolacha* → bo'la.ʃa → bu'la.ʃɐ vs. *bo'la.ʃɐ). Nevertheless, the mappings [+low] → [−low] and [−high] → [+high] do not occur simultaneously; rather, only the former takes place.

From (44), the following constraint rankings can be inferred:

*[+low]/V[−stressed, μ] >> IDENT(low)

*[−high]/V[−stressed, μ] >> IDENT(high)

*[+low]/V[−stressed, μ] >> *[−high]/V[−stressed, μ]

The definitions of the above constraints are as follows:

*[+low]/V[−stressed, μ]: Assign one * for each unstressed low short vowel.

*[−high]/V[−stressed, μ]: Assign one * for each unstressed non-high short vowel.

IDENT(low): Assign one * for each change in the [low] feature between input and output.

IDENT(high): Assign one * for each change in the [high] feature between input and output.

However, the above constraint ranking alone cannot prevent a candidate that simultaneously undergoes the mappings [+low] → [−low] and [−high] → [+high] — such as *bolacha* bo'la.ʃa → *bu'la.ʃi — from incorrectly winning over the expected output bu'la.ʃɐ. To resolve this, a higher-ranked faithfulness constraint must be introduced above the ranking *[+low]/V[−stressed, μ] >> *[−high]/V[−stressed, μ]. This constraint should penalise the simultaneous violations of both IDENT(low) and IDENT(high) — as in the mapping a → *i — while still favouring the mapping a → ɐ, which only violates IDENT(low). In doing so, the optimal candidate ɐ will correctly emerge as the winner over *i. The required constraint is therefore one that **allows for a change in either the [+low] or [−high] feature independently, but prohibits simultaneous changes in both**. This constraint can be formalised as:

IDENT(lowVhigh): Assign one * if both IDENT(low) and IDENT(high) are violated in a single

mapping.

This constraint is crucial for ensuring that /a/ maintains its distinctive opposition to /e ε/, which surface as [i]. In this sense, its introduction represents **a theoretical expression of a functional requirement within the EP system.** From this, the following rankings are inferred:

IDENT(lowVhigh) >> *[+low]/V[–stressed, μ] >> IDENT(low)

IDENT(lowVhigh) >> *[+low]/V[–stressed, μ] >> *[–high]/V[–stressed, μ] >> IDENT(high)

Of these, the ranking relationships between IDENT(low) and *[–high]/V[–stressed, μ], as well as between IDENT(low) and IDENT(high), remain undetermined. The reasons are as follows:

The winning candidates supported by IDENT(low) — namely, *i e ε ɔ o u* → *i i i u u u* — are all [+high], and thus also supported by the markedness constraint *[–high]/V[–stressed, μ]. As a result, these two constraints do not conflict.

As for IDENT(low) and IDENT(high), there are cases where the winning candidate is favoured by IDENT(low), and others where it is favoured by IDENT(high). In either case, however, the candidate is also supported by a higher-ranked constraint, and thus no conflict between IDENT(low) and IDENT(high) becomes decisive in the evaluation.

Now from the data in (42), the descriptive generalisation (45) can be formulated:

(45) In unstressed syllables, short vowels that stand in opposition with respect to [ATR] are prohibited. This requirement is enforced by the unfaithful mapping $[\pm ATR] \rightarrow [0 ATR]$, as in *morada* "address" /mɔr-ad-a/ → mɔ'ra.da → mu'ra.dɐ vs. *mɔ'ra.dɐ, *mɔ'ra.dɐ, *mɔ'ra.dɐ.

From this description, the following constraint hierarchies can also be posited:

***[±ATR]/V[–stressed, μ] >> IDENT(ATR)**

The definitions of the above constraints are as follows:

***[±ATR]/V[–stressed, μ]:** Assign one * for each unstressed short vowel that is specified with respect to [ATR].

IDENT(ATR): Assign one * for each change in the [ATR] feature between input and output.

The following tableaux (46) and (47) serve to confirm the ranking proposed above (note that the contextual specification "/V[–stressed, μ]" is omitted in the tableaux due to space constraints):

(46)	a	ID(lowV/high)	*[+low]	*[−high]	ID(high)	*[+low]	ID(low)	*[±ATR]	ID(ATR)
	ɛ			*			*		*
	a		*W	*		*W	L		*
	ɜ			*			*	*W	L
	i	*W		L	*W		*		*

(47)	ɔ	ID(lowV/high)	*[+low]	*[−high]	ID(high)	*[+low]	ID(low)	*[±ATR]	ID(ATR)
	ɛ				*				*
	ɔ			*W	L			*W	L
	o			*W	L			*W	*
	u				*			*W	L

Finally, from the data in (43), the descriptive generalisation (48) can be formulated:

(48) In unstressed syllables, short front vowels specified as [−back] are prohibited. This requirement is enforced by an unfaithful mapping from [−back] to [+back] — e.g. *dedada* "fingerprint" /ded-ad-a/ → de'da.da → di'dadə vs. *dedada, *di'dadə, and *pegado* "next-door" /pɛg-a-d-o/ → pɛ'ga.do → pi'ga.du vs. *pɛ'ga.du, *pe'ga.du, *pi'ga.du.

From this generalisation, the following ranking can be inferred:

***[−back]/V[−stressed, μ] >> IDENT(back)**

The definitions of these constraints are as follows:

***[−back]/V[−stressed, μ]**: Assign one * for each unstressed short front vowels.

IDENT(back): Assign one * for each change in the [back] feature between input and output.

The following tableau (49) serves to confirm the ranking proposed just above:

(49)	e	*[−back]/V[−stressed, μ]	IDENT(back)
	ɛ		*
	i	*W	L

However, **the mapping [−back] → [+back] always occurs in tandem with the mapping [−high] → [+high] and never in isolation** — that is, /e ɛ/ → [i] vs. *[i], whereas /i/ → [i] vs. *[i]. Therefore, if we evaluate the output candidates [i] and *[i] for the input /i/, based on the previously established ranking ***[−back]/V [−stressed, μ] >> IDENT(back)**, the candidate *[i], which is favoured by the higher-ranked ***[−back]/V[−stressed, μ]**, would incorrectly win over

the optimal candidate [i], which is favoured by the lower-ranked IDENT(back), as shown in the tableau (50). This contradicts the observed facts:

(50)

i	*[−back]/V[−stressed, μ]	IDENT(back)
i	*	
i	L	*W

If we reverse, however, the ranking between *[−back]/V[−stressed, μ] and IDENT(back), adopting IDENT(back) >> *[−back]/V[−stressed, μ], we can no longer account for the mapping /e/ → [i] vs. *[i]. Thus, **to derive the correct outputs, it is necessary to maintain the ranking *[−back]/V[−stressed, μ] >> IDENT(back), while positing an additional constraint ranked above *[−back]/V[−stressed, μ] that favours /i/ → [i] and disfavors /i/ → *[i].**

Among the output candidates for /i/, [i] and *[i] differ only in the value of the [back] feature — the former being [−back], faithful to the input, and the latter [+back], unfaithful to the input — while they share identical values for all the remaining features: [+high], [−low], [+ATR] and [−round]. Therefore, no markedness constraint pertaining to those shared features, even if ranked above *[−back]/V[−stressed, μ], can favour [i] over *[i]: such constraints would support neither candidate. In fact, among all the markedness constraints extracted thus far, only *[−back]/V[−stressed, μ] expresses any preference between [i] and *[i]; the remaining constraints — *[+low]/V[−stressed, μ], *[−high]/V[−stressed, μ] and *[−ATR]/V[−stressed, μ] — favour neither. Even the introduction of additional markedness constraints such as *[−round] would yield the same result, as previously noted.

It is reasonable, therefore, to assume that **the constraint ranked above *[−back]/V[−stressed, μ] must be some faithfulness constraint**. Until its specific content is determined, we shall refer to this constraint provisionally as *F*. When this ranking — *F* >> *[−back]/V[−stressed, μ] >> IDENT(back) — is incorporated into the tableaux in (49) and (50), it yields the following constraint interactions in (51) and (52) (note that no * is assigned in the column for *F*, as its definition remains unspecified):

(51)

e	<i>F</i>	*[−back]/V[−stressed, μ]	IDENT(back)
i			*
i	(W)	*W	L

(52)

i	<i>F</i>	*[−back]/V[−stressed, μ]	IDENT(back)
i		*	
i	W	L	*W

The faithfulness constraint *F* must satisfy the following two conditions within the ranking $F \gg$ *[back]/V[−stressed, μ] $\gg$ IDENT(back):

(53) *F* must **favour the mapping /i/ → [i] while disallowing /i/ → *[i]**.

(54) *F* must not disallow the mapping /e/ → [i]; that is, it must either **be neutral with respect to both /e/ → [i] and /e/ → *[i]**, or else actively **favour /e/ → [i]**.

Let us first consider the faithfulness violations involved in the mapping of /i/, with respect to condition (53). The mapping /i/ → [i] incurs no violations, whereas /i/ → *[i] involves a single faithfulness violation: the change from [−back] to [+back]. Therefore, the faithfulness constraint *F* can be taken as one that **prohibits the mapping [−back] → [+back], thereby disfavoring /i/ → *[i]**. However, since this constraint must outrank *[−back]/V[−stressed, μ], it is **evidently not identical to IDENT(back)**, given the ranking *[−back]/V[−stressed, μ] $\gg$ IDENT(back).

Next, let us examine the faithfulness violations involved in the mappings of /e/, with respect to condition (54). The mapping /e/ → *[i] incurs a single violation: [−high] → [+high]. On the other hand, /e/ → [i] incurs two violations: [−high] → [+high] and [−back] → [+back]. That is, **/e/ → [i] entails more faithfulness violations than /e/ → *[i]**. It is therefore **unlikely that a faithfulness constraint *F* could favour the more unfaithful /e/ → [i] while disallowing the less unfaithful /e/ → *[i]**. Consequently, the most plausible assumption is that **the faithfulness constraint *F* supports neither /e/ → [i] nor /e/ → *[i]**.

Moreover, if the faithfulness constraint *F* were to target only the mapping [−back] → [+back], then *F* would effectively be IDENT(back), which would contradict our assumption that *F* must outrank *[−back]/V[−stressed, μ], given the ranking *[−back]/V[−stressed, μ] $\gg$ IDENT(back). Therefore, **the faithfulness constraint *F* must also be concerned with the concomitant mapping [−high] → [+high] found in both /e/ → [i] and /e/ → *[i], and likely serves to prohibit that change as well.**

To summarise the above discussion, **the faithfulness constraint *F* in question must simultaneously prohibit both the change [−back] → [+back] and [−high] → [+high]**. In order for such a constraint to disfavour the mapping /i/ → *[i] in favour of /i/ → [i], while assigning the same number of violations to both /e/ → [i] and /e/ → *[i], we can define *F* as

IDENT(high^back). The definition of this constraint is as follows:

IDENT(high^back): Assign one * if either IDENT(high) or IDENT(back) is violated in a single mapping.

By ranking it as follows:

IDENT(high^back) >> *[-back]/V[-stressed, μ] >> IDENT(back)

we can coherently account for both the mapping /i/ → [i] vs. *[i] and the mapping /e/ → [i] vs. *[i] without any contradiction. Put simply, within the [-low] domain defined by [±high] and [±back], any deviation from the input — whether a single step (as in /i/ → [i] or /e ε/ → [i]) or a double step (as in /e ε/ → [i] → [i]) — results in an equal degree of violation. And when the same violations are incurred, the candidate that achieves a more optimal shape — namely, /e ε/ → [i] → [i] — emerges as the winner. The following tableaux (55) and (56) serve to confirm the ranking inferred above:

(55)

e	IDENT(high^back)	*[-back]/V[-stressed, μ]	IDENT(back)
$\text{[e]}\rightarrow\text{[i]}$	*		*
i	*	*W	L

(56)

i	IDENT(high^back)	*[-back]/V[-stressed, μ]	IDENT(back)
$\text{[e]}\rightarrow\text{[i]}$		*	
i	*W	L	*W

The introduction of the complex constraint **IDENT(high^back)** here — like that of **IDENT(low^high)** above — serves not merely as a technical expedient, but also as a **theoretical expression of a functional requirement within the EP system: the maintenance of the distinctive opposition between the underlying high front vowel /i/ and the underlying mid front vowels /e ε/**. Thus, the ranking of **IDENT(high^back) >> [-back]/V[-stressed, μ]** should be interpreted as reflecting the system's resistance to neutralising this distinctive opposition. Without such a constraint, the system would favour maximal reduction to a reduced vowel [i] across the entire front vowel space /i e ε/, thereby collapsing an opposition that is both frequent and functionally load-bearing.

While the general vowel reduction pattern in unstressed syllables preserves the underlying opposition between /i/ and /e ε/, realised phonetically by the opposition between [i] and [i],

through the ranking $\text{IDENT}(\text{high}\wedge\text{back}) \gg [-\text{back}]/\text{V}[-\text{stressed}, \mu] \gg \text{IDENT}(\text{back})$, a more neutralised pattern is observed in the prosodically weakest final syllables. Specifically, in final unstressed open syllables or in final unstressed syllables closed by [ʃ] or [ʒ] — the only consonants specified as $[-\text{sonorant}]$ in syllable coda — even /i/ surfaces as [i], as shown in (57):

$$(57) \text{part}[\acute{\text{i}}:] \text{mos} \approx \text{part}[\text{i}]; \text{com}[\acute{\text{e}}:] \text{mos} \approx \text{com}[\text{i}]; \text{diss}[\acute{\text{e}}:] \text{mos} \approx \text{diss}[\text{i}]$$

(Note that *partimos*, *parte* and *comemos*, *come* are the first person plural and third person singular of the verbs *partir* "to depart" and *comer* "to eat" in the present indicative, respectively; *dissemos* and *disse* are the first person plural and third person singular of *dizer* "to say" in the imperfect subjunctive.) This neutralisation can be accounted for by a positional markedness constraint $*[-\text{back}]/\text{V}[-\text{stressed}, \mu] __ ([-\text{sonorant}])\#$, which dominates $\text{IDENT}(\text{high}\wedge\text{back})$. The definition of this constraint is as follows:

$*[-\text{back}]/\text{V}[-\text{stressed}, \mu] __ ([-\text{sonorant}])\#$: Assign one * for each unstressed short front vowel that occurs in a final open syllable or in a final syllable closed by a $[-\text{sonorant}]$ segment.

The implication of this constraint is that, **in the weakest position at the word-final edge, the underlying opposition between /i/ and /e ε/ can no longer be maintained**, as seen in /i e ε a ɔ o u/ → [i i i ɐ u u u]. In "Brazil where the posttonic syllables...are characteristically weak" (Mattoso Câmara 1972: 26), the opposition between /i/ and /e ε/ is likewise neutralised in this same weakest position, yielding /i e ε a ɔ o u/ → [i i i ɐ u u u]; in other unstressed positions, by contrast, the opposition is preserved, as seen in /i e ε a ɔ o u/ → [i e e a o o u].

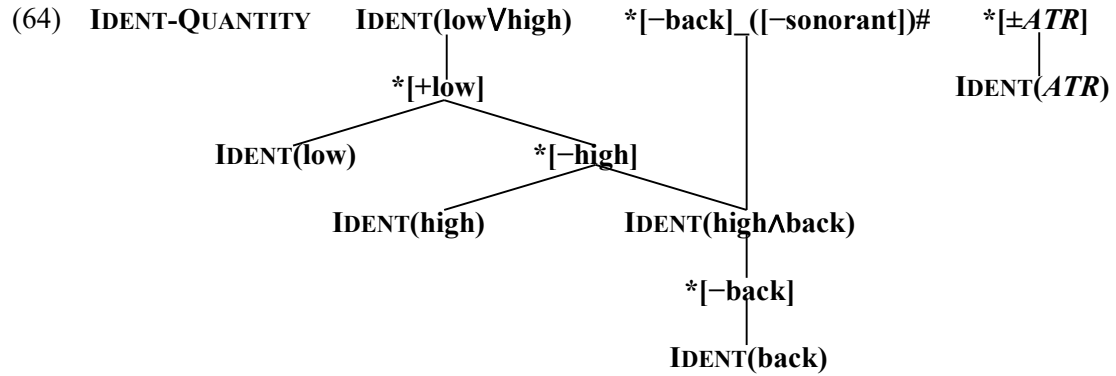
Now, since $\text{IDENT}(\text{high}\wedge\text{back})$ is a faithfulness constraint that is related not only to [back] but also to [high], it comes into conflict not only with markedness constraints applying to the former — such as $[-\text{back}]/\text{V}[-\text{stressed}, \mu]$ and $[\text{back}]/\text{V}[-\text{stressed}, \mu] __ ([-\text{sonorant}])\#$ — but also with the markedness constraint $[-\text{high}]/\text{V}[-\text{stressed}, \mu]$, which applies to the latter. For example, in the mapping $e \rightarrow i$ vs. $*\epsilon$ — e.g. *dedada* /ded-ad-a/ → de'da.da → di'da.dɐ vs. $*d\epsilon$ 'da.dɐ — the following patterns of constraint satisfaction and violation are observed:

$e \rightarrow i$: obeys $*[-\text{high}]/\text{V}[-\text{stressed}, \mu]$, $*[\pm\text{ATR}]/\text{V}[-\text{stressed}, \mu]$, $*[-\text{back}]/\text{V}[-\text{stressed}, \mu]$;

violates $\text{IDENT}(\text{high})$, $\text{IDENT}(\text{ATR})$, $\text{IDENT}(\text{high}\wedge\text{back})$, $\text{IDENT}(\text{back})$

$e \rightarrow *\epsilon$: obeys $\text{IDENT}(\text{high})$, $*[\pm\text{ATR}]/\text{V}[-\text{stressed}, \mu]$, $\text{IDENT}(\text{high}\wedge\text{back})$, $\text{IDENT}(\text{back})$;

violates $*[-\text{high}]/\text{V}[-\text{stressed}, \mu]$, $\text{IDENT}(\text{ATR})$, $*[-\text{back}]/\text{V}[-\text{stressed}, \mu]$



This was not intended at all, but it is noteworthy that the constraint hierarchy $*[+low]/V[-stressed, \mu] \gg *[-high]/V[-stressed, \mu] \gg *[-back]/V[-stressed, \mu]$ in (64) coincides with the logical process of vowel reduction in unstressed syllables analysed in (33)-(35), which suggests that the constraint hierarchy reflects the functioning of the system.

The following tableaux (65) - (71) serve to test the constraint ranking established in (64):

(65) *bolacha* /bol-af-a/

bo'la.fa	bu'la.fɐ	bu'la.fɔ	bu'la.fɪ	bu'la.fɐ	bo'la.fɐ
IDENT(lowVhigh)			*W		
*[+low]		*W			
*[-high]	*	*	L	*	**W
IDENT(high\back)	*	*	**W	*	L
*[-back]					
IDENT(back)					

IDENT(lowVhigh)			*W		
*[+low]		*W			
IDENT(low)	*	L	*	*	*

IDENT(lowVhigh)			*W		
*[+low]		*W			
*[-high]	*	*	L	*	**W
IDENT(high)	*	*	**W	*	L

*[-back]_([-sonorant])#					
IDENT(high\back)	*	*	**W	*	L
*[-back]					
IDENT(back)					

*[\pmATR]					*W
IDENT(ATR)	**	**	**	**	*L

(66) *morada* /mɔr-ad-a/

mɔ'ra.da	mu'ra.dɐ	mɔ'ra.dɐ	mɔ'ra.dɐ	mɔ'ra.dɐ
IDENT(lowVhigh)				
*[+low]				
*[−high]	*	**W	**W	*
IDENT(high^back)	*	L	L	*
*[−back]				
IDENT(back)				

IDENT(lowVhigh)				
*[+low]				
IDENT(low)	*	*	*	*

IDENT(lowVhigh)				
*[+low]				
*[−high]	*	**W	**W	*
IDENT(high)	*	L	L	*

*[−back]_([−sonorant])#				
IDENT(high^back)				
*[−back]				
IDENT(back)				

*[±ATR]		*W	*W	*W
IDENT(ATR)	*	L	*	L

(67) *pegado* /pɛg-a-d-o/

pɛ'ga.do	pi'ga.du	pɛ'ga.du	pɛ'ga.du	pi'ga.du
IDENT(lowVhigh)				
*[+low]				
*[−high]		*W	*W	
IDENT(high^back)	**	*L	*L	**
*[−back]		*W	*W	*W
IDENT(back)	*	L	L	L

IDENT(lowVhigh)				
*[+low]				
IDENT(low)				

IDENT(lowVhigh)				
*[+low]				
*[−high]		*W	*W	
IDENT(high)	**	*L	*L	**

*[−back]_([−sonorant])#				
IDENT(high^back)	**	*L	*L	**
*[−back]		*W	*W	*W
IDENT(back)	*	L	L	L

*[±ATR]		*W	*W	
IDENT(ATR)	**	*L	**	**

(68) *cinema* /sinemat/

si'ne.ma	si'neme	si'ne.mə
IDENT(lowVhigh)		
*[+low]		
*[-high]	*	*
IDENT(high^back)		*W
*[-back]	*	L
IDENT(back)		*W

IDENT(lowVhigh)		
*[+low]		
IDENT(low)	*	*

IDENT(lowVhigh)		
*[+low]		
*[-high]	*	*
IDENT(high)		

*[-back]_([-sonorant])#		
IDENT(high^back)		*W
*[-back]	*	L
IDENT(back)		*W

*[±ATR]		
IDENT(ATR)	**	**

(69) *parte* /part-i/

'par.ti	'par.ti	'par.ti
IDENT(lowVhigh)		
*[+low]		
*[-high]		
IDENT(high^back)	*	L
*[-back]		*W
IDENT(back)	*	L

IDENT(lowVhigh)		
*[+low]		
IDENT(low)		

IDENT(lowVhigh)		
*[+low]		
*[-high]		
IDENT(high)		

*[-back]_([-sonorant])#		*W
IDENT(high^back)	*	L
*[-back]		*W
IDENT(back)	*	L

*[±ATR]		
IDENT(ATR)	*	*

(70) *come* /kɔm-e/

'kɔ.me	ᑭᓐᑭᓐ 'kɔ.mi	'kɔ.me	'kɔ.mi
IDENT(lowVhigh)			
*[+low]			
*[-high]		*W	
IDENT(high^back)	*	L	*
*[-back]		*W	*W
IDENT(back)	*	L	L

IDENT(lowVhigh)			
*[+low]			
IDENT(low)			

IDENT(lowVhigh)			
*[+low]			
*[-high]		*W	
IDENT(high)	*	L	*

*[-back]_([-sonorant])#		*W	*W
IDENT(high^back)	*	L	*
*[-back]		*W	*W
IDENT(back)	*	L	L

*[±ATR]		*W	
IDENT(ATR)	*	L	*

(71) *disse* /dis-ɛ/

'di.sɛ	ᑭᓐᑭᓐ 'di.si	'di.sɛ	'di.se	'di.si
IDENT(lowVhigh)				
*[+low]				
*[-high]		*W	*W	
IDENT(high^back)	*	L	L	*
*[-back]		*W	*W	*W
IDENT(back)	*	L	L	L

IDENT(lowVhigh)				
*[+low]				
IDENT(low)				

IDENT(lowVhigh)				
*[+low]				
*[-high]		*W	*W	
IDENT(high)	*	L	L	*

*[-back]_([-sonorant])#		*W	*W	*W
IDENT(high^back)	*	L	L	*
*[-back]		*W	*W	*W
IDENT(back)	*	L	L	L

*[±ATR]		*W	*W	
IDENT(ATR)	*	L	*	*

4.2. Vocalic Quantity Adjustment Stratum

From (23), it was assumed that major vowel quality changes occur at the Vocalic Quality Modification Stratum (24), and that only quantity adjustment takes place at this Vocalic Quantity Adjustment Stratum, as shown in (25). Accordingly, IDENT-QUALITY is assumed to be an inviolable constraint operating at this stratum. The definition of this constraint is as follows:

IDENT-QUALITY: Assign one * for each change in the value of any quality feature — e.g. [low], [high], [ATR], [back], [round] — between input and output.

Turning to vowel quantity alternations in Contemporary EP, the following descriptive generalisation may be formulated as (72):

(72) Long vowels are prohibited. This requirement is enforced by deleting one mora if the input is a long vowel (e.g. *ativo* "active" /a:t-iv-o/ → a:'ti.vu → a'ti:.vu vs. *a:'ti:.vu), suggesting the ranking *V[μμ] >> MAX(μ),

except when the vowel occurs in a stressed syllable, where short vowels are prohibited. This requirement is enforced by epenthesising one mora if the input is a short vowel (e.g. *ativo* "active" /a:t-iv-o/ → a:'ti.vu → a'ti:.vu vs. *a'ti.vu; *pasta* "file-folder" /past-a/ → 'paʃ.tə → 'pa:ʃ.tə vs. *'paʃ.tə), suggesting the ranking *'σ[μ] >> DEP(μ), or by maintaining the quantity if the input is a long vowel (e.g. *ata* "minutes, record" /a:t-a/ → 'a:.tə → 'a:.tə vs. *'a.tə; *mestre* "master" /mɛ:stɹ-e/ → 'mɛ:ʃ.tri → 'mɛ:ʃ.tri vs. *'mɛʃ.tri), suggesting the ranking *'σ[μ] >> *V[μμ],

except when the vowel occurs in a syllable ended by a moraic [+sonorant] segment — namely, [ɬ], [ɾ], [ɪ] or [ʊ] — where long vowels are prohibited. This requirement is enforced by deleting one mora if the input is a long vowel (e.g. *norma* "regulation, norm" /nɔ:rm-a/ → 'nɔ:ɾ.mə → 'nɔɾ.mə vs. *'nɔ:ɾ.mə; *normal* "normal" /nɔ:rm-al-0/ → nɔ:ɾ'ma.ɪ → nɔɾ'ma:ɪ vs. *nɔ:ɾ'ma:ɪ), suggesting the ranking *σ[μμμ] >> *'σ[μ]⁹.

The definitions of the above constraints are as follows:

*σ[μμμ]: Assign one * for each syllable with three moras.

*'σ[μ]: Assign one * for each stressed syllable with only one mora.

*V[μμ]: Assign one * for each vowel linked to two moras.

⁹ The radical vowel -o- in *n[ɔ]rma* ≈ *n[ɔ]rmaɪ*, though possibly a short vowel etymologically, is analysed here as an underlying long vowel, since in the present state of language it does not alternate in quality — unlike the -o- in *f[ɔ]rma* "form" ≈ *f[u]rmaɪ* "formal".

MAX(μ): Assign one * for each mora deleted.

DEP(μ): Assign one * for each mora epenthesised.

From the descriptive generalisation (72), the constraint rankings (73) and (74) may be inferred:

(73) $*\sigma[\mu\mu\mu] \gg *'\sigma[\mu] \gg *V[\mu\mu] \gg \text{MAX}(\mu)$

(74) $*\sigma[\mu\mu\mu] \gg *'\sigma[\mu] \gg *V[\mu\mu], \text{DEP}(\mu)$

On the other hand, syllable/vowel quantity adjustment can also be achieved not only through moraic operations but also by the epenthesis or deletion of segments. For example, in mappings such as *ata* (noun) /a:ta/ $\rightarrow$ 'a:.tə $\rightarrow$ 'a:.tə vs. *'ar.tə, or *norma* /nɔ:rma/ $\rightarrow$ 'nɔ:r.mə $\rightarrow$ 'nɔr.mə vs. *'nɔ:.mə, the asterisked candidates *'ar.tə and *'nɔ:.mə are eliminated due to violations of DEP and MAX, respectively. The definitions of these constraints are as follows (here "segment" refers to a root node in the quality feature geometry, abstracting away from quantity; therefore, deletion or insertion of moras alone is not to be considered a violation of DEP or MAX):

DEP: Assign one * for each segment epenthesised.

MAX: Assign one * for each segment deleted.

First, in the mapping *ata* /a:t-a/ $\rightarrow$ 'a:.tə $\rightarrow$ 'a:.tə vs. *'ar.tə, the winner 'a:.tə is favoured by DEP, while the loser *'ar.tə is favoured by $*V[\mu\mu]$, which indicates the ranking **DEP** $\gg$ $*V[\mu\mu]$. Note that both candidates violate none of the constraints $*\sigma[\mu\mu\mu]$, $*'\sigma[\mu]$, MAX, MAX(μ) and DEP(μ), and thus are equally evaluated with respect to these constraints. Second, in the mapping /a:t-a/ $\rightarrow$ 'a:.tə $\rightarrow$ 'a:.tə vs. *tə, the winning candidate 'a:.tə is favoured by MAX, whereas the losing candidate *tə is favoured by $*V[\mu\mu]$, which supports the ranking **MAX** $\gg$ $*V[\mu\mu]$. Note also that 'a:.tə satisfies MAX(μ), while *[tə] violates it; given the ranking $*V[\mu\mu] \gg \text{MAX}(\mu)$ in (73), however, the candidate evaluation by MAX(μ) does not affect the evaluation determined by $*V[\mu\mu]$. Since MAX and DEP do not conflict in syllable/vowel quantity adjustment, the following constraint ranking (75) can result from the above discussion:

(75) **MAX, DEP** $\gg$ $*V[\mu\mu]$

Next is the ranking among MAX, DEP and $*\sigma[\mu\mu\mu]$, all of which are ranked above $*V[\mu\mu]$. With the exception of word-final trimoraic syllables — such as [-'me:r] in *comer* /kɔm-e-r0/ $\rightarrow$ [ku'me:r] "to eat" — which require special theoretical treatment and which will be discussed below, trimoraic syllables are found only in loanwords, as in *e-mail* "e-mail" /imeil/ $\rightarrow$ i'meɪɭ ($\rightarrow$ [i'meɪɭ]), with the final syllable -'meɪɭ. In such cases, neither epenthesis, as in *i.meɪ.li ($\rightarrow$

*[i'mɛj.li]), nor deletion, as in *i'meĭ (→ *[i'mɛj]), is attested, which suggests, at least provisionally, the ranking in (76):

(76) MAX, DEP ≫ *σ[μμμ]

Now let us turn to the diphthongisation of the stressed short vowels /e/ and /ɛ/ in hiatus. As stated in (72), short vowels in stressed CV syllables surface as long vowels through the addition of one mora. However, when such an open syllable is immediately followed by a vowel-initial syllable — that is, when the stressed vowel and the following unstressed vowel form a hiatus — the situation becomes somewhat complex. The short vowels /i/, /u/ and /o/ undergo lengthening with no change in quality: *tia* "aunt" /ti-a/ → 'ti.a → 'ti.ɐ → ['ti:.ɐ]; *tua* "your (fem.)" /tu-a/ → 'tu.a → 'tu.ɐ → ['tu:.ɐ]; *boa* "good (fem.)" /bo-a/ → 'bo.a → 'bo.ɐ → ['bo:.ɐ].

On the other hand, /e/ and /ɛ/, specified as [–low, –high, –back], undergo both lengthening and concomitant diphthongisation in ĭ — or epenthesis of ĭ immediately following them — e.g. *teia* "cobweb" /te-a/ → 'te.a → 'te.ɐ → 'teĭ.ɐ; *ateia* "atheist" /a-tɛ-a/ → a'tɛ.a → ɐ'tɛ.ɐ → ɐ'teĭ.ɐ. (Subsequently, eĭ and ɛĭ dissimilate to [ɐĭ] at a later stratum, which produces the final surface forms *teia* ['teĭ.ɐ] and *ateia* [ɐ'teĭ.ɐ] in EP, whereas in BP this dissimilation does not take place and eĭ and ɛĭ surface as [eĭ] and [ɛĭ], as in *teia* ['teĭ.ɐ] and *ateia* [a'tɛĭ.ɐ]. In this stratum, a markedness constraint — tentatively named OCP(back)/Rhyme — that prohibits adjacent [αback] (where α is a variable) within the same rhyme, as in *[eĭ], *[ɛĭ] and *[ou], when properly ranked, might account for the exclusion of ill-formed outputs in EP such as *['teĭ.ɐ], *[ɐ'teĭ.ɐ] and *douto* "erudite" /dout-o/ → *['dou.tu] vs. ['do:.tu], for example.)

The next issue is how to analyse the changes –é.ɐ → –éĭ.ɐ and –ɛ.ɐ → –ɛĭ.ɐ. It is possible, as was suggested above, to interpret this change not as the epenthesis of a segmental ĭ between e or ɛ and ɐ, but rather as the result of lengthening e and ɛ by one mora, during which the features [+high] and [0ATR] are sequentially added following the original [–high] and [+ATR] or [–ATR] specifications within the internal feature geometry of e and ɛ, as shown in (77), where the added feature values are shaded:

(77)

	^μ e		^{μμ} eĭ		^μ ɛ		^{μμ} ɛĭ
low	–		–		–		–
high	–		– +		–		– +
ATR	+	⇒	+ 0		–	⇒	– 0
back	–		–		–		–
round	–		–		–		–

This analysis is conceptually analogous to a geometrical analysis of the affrication of plosives, where [+continuant] is sequentially added at the root node after [–continuant]. From the above discussion, the following descriptive generalisation (78) may be formulated:

(78) In a stressed open syllable before another syllable initiated by a vowel, short vowels are prohibited. This requirement is enforced by simply epenthesising one mora if the input is a short vowel (e.g. *tia* /ti-a/ → 'ti.a → 'ti.ə → 'ti:ə vs. *ti.ə; *tua* /tu-a/ → 'tu.a → 'tu.ə → 'tu:ə vs. *tu.ə; *boa* /bo-a/ → 'bo.a → 'bo.ə → 'bo:ə vs. *bo.ə), suggesting the ranking *'σ[μ] >> DEP(μ) as in (72),

except when a long mid front vowel — namely, é: or é: in this language — would result, in which case [+high] and [0ATR] are added sequentially after the original [–high] and [+ATR] or [–ATR] within the feature geometry of e and ε (e.g. *teia* /te-a/ → 'te.a → 'te.ə → 'tej.ə vs. *te.ə, *te:ə; *ateia* /a-tə-a/ → a'tə.a → ə'tə.ə → ə'tej.ə vs. *ə'tə.ə, *ə'te:ə), suggesting the rankings *V[–low, –high, –back, μμ]_V >> DEP(high), DEP(ATR) and *'σ[μ] >> DEP(high), DEP(ATR),

furthermore, the off-glide corresponding to the epenthesised mora is prohibited from being [–high, ±ATR] (e.g. *ateia* /a-tə-a/ → a'tə.a → ə'tə.ə → ə'tej.ə vs. *ə'tə.ə, *ə'tej.ə), suggesting the ranking *σ[C₀V₁V₂[–high, ±ATR]] >> DEP(high), DEP(ATR).

The definitions of the above constraints are as follows:

*V[–low, –high, –back, μμ]_V: Assign one * for each long mid front vowel occurring before another vowel in the following syllable.

DEP(high): Assign one * for each instance of [high] that is newly inserted into the original feature geometry of a segment.

DEP(ATR): Assign one * for each instance of [ATR] that is newly inserted into the original feature geometry of a segment.

*σ[C₀V₁V₂[–high, ±ATR]]: Assign one * for each vocoid specified either as [–high] or as [±ATR] that occurs as an off-glide within the same nucleus.

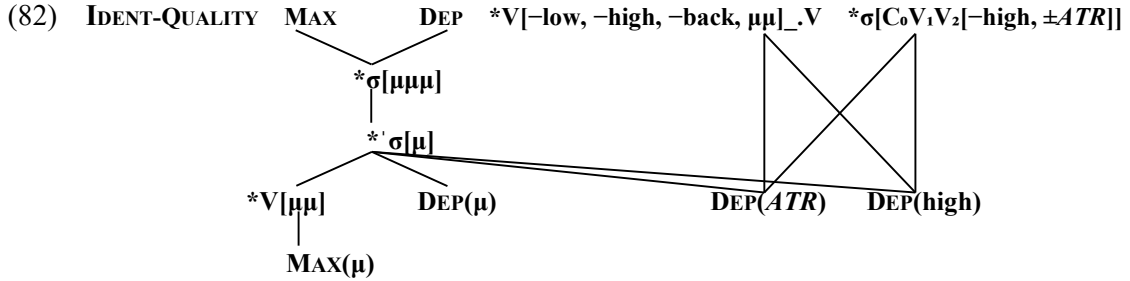
From the descriptive generalisation (78), the constraint rankings (79) - (81) may be inferred:

(79) *V[–low, –high, –back, μμ]_V >> DEP(high), DEP(ATR)

(80) *'σ[μ] >> DEP(μ), DEP(high), DEP(ATR)

(81) $*\sigma[C_0V_1V_2[-\text{high}, \pm\text{ATR}]] \gg \text{DEP}(\text{high}), \text{DEP}(\text{ATR})$

The Hasse diagrams for the constraint hierarchies in (73) - (81) are as shown in (82):



Let us return to the trimoraic $[-'me:r]$ in *comer* $[ku'me:r]$. Assuming that the unmarked stress assignment in EP — which we hypothetically take to be based on trochaic foot construction starting from the word-final syllable — also applies to impersonal infinitives such as *falar*, *comer* and *partir*, the underlying representation of *comer* is most likely $/k\text{om-e-r}0/$, where $/k\text{om-}/$ constitutes the verbal radical, $/-e/$ the thematic vowel and $/-r0/$ the infinitive suffix. In this representation, $/0/$ denotes a vowel with no segmental quality but bearing a vocalic quantity of one mora. When syllabified, this underlying representation is assumed to take the form $k\text{om.me.r}$, with the consonant $/r/$ linked to the floating mora of $/0/$, which receives main stress on the syllable $-me-$, yielding the derived form $k\text{om}'me.r$. This representation, after undergoing lenition of r to ɾ prior to vowel nasalisation — that is, $k\text{om}'me.\text{ɾ}$ — yields at the Vocalic Quality Modification Stratum the form $ku'me.\text{ɾ}$, which subsequently undergoes quantity adjustment at the Vocalic Quantity Adjustment Stratum, producing the output $ku'me:.\text{ɾ}$. This process is summarised in (83), where the derivation of *comera* $/k\text{om-e-ra}/$ — the first/third person singular of the verb *comer* in the pluperfect indicative — is provided for reference:

(83) *comer* $/k\text{om-e-r}0/ \rightarrow k\text{om.me.r} \rightarrow k\text{om}'me.\text{ɾ} \rightarrow ku'me.\text{ɾ} \rightarrow ku'me:.\text{ɾ} (\rightarrow [ku'me:r])$

comera $/k\text{om-e-ra}/ \rightarrow k\text{om.me.ra} \rightarrow k\text{om}'me.ra \rightarrow ku'me.r\text{e} \rightarrow ku'me:.\text{r\text{e}} (\rightarrow [ku'me:.\text{r\text{e}}])$

Subsequently, resyllabification in a later stratum — where, for instance, the diphthongs $e\text{ɪ}$ and $e\text{ɪ}$ undergo adjacent dissimilation, changing to $[e\text{ɪ}]$, as has already been stated — produces the final surface form $[ku'me:r]$. In this stratum, the ranking $*C, \text{MAX}(\mu), \text{MAX}, \text{DEP} \gg *\sigma[\mu\mu\mu]$ appear to account for the exclusion of ill-formed candidates such as $*[ku'mer]$, $*[ku'me:r\text{ɪ}]$ and $*[ku'me:]$. Under this analysis, therefore, the output of $/k\text{om-e-r}0/$ at the Vocalic Quantity Adjustment Stratum is $ku'me:.\text{ɾ}$, which does not violate the constraint $*\sigma[\mu\mu\mu]$.

The following tableaux (84) - (93) serve to test the constraint ranking established as in (82):

(84) *ativo* /a:t-iv-o/

a:ˈti.vu	ID-QUAL	MAX	DEP	*σ[μμμ]	*'σ[μ]	*V[μμ]	MAX(μ)	*'σ[μ]	DEP(μ)
☞ aˈtiː.vu						*	*		*
a:ˈtiː.vu						**W	L		*
aˈti.vu					*W	L	*	*W	L
a:ˈti.vu					*W	*	L	*W	L

(85) *pasta* /pasta/

ˈpas.tə	ID-QUAL	MAX	DEP	*σ[μμμ]	*'σ[μ]	*V[μμ]	MAX(μ)	*'σ[μ]	DEP(μ)
☞ ˈpaːʃ.tə						*			*
ˈpaʃ.tə					*W	L		*W	L

(86) *ata* /a:t-a/ (noun)

ˈaː.tə	ID-QUAL	MAX	DEP	*σ[μμμ]	*'σ[μ]	*V[μμ]	MAX(μ)	*'σ[μ]	DEP(μ)
☞ ˈaː.tə						*			
ˈa.tə					*W	L	*W	*W	
ˈaɾ.tə			*W			L			
tə		*W				L	**W		

(87) *mestre* /mɛːstr-e/

ˈmɛːʃ.tri	ID-QUAL	MAX	DEP	*σ[μμμ]	*'σ[μ]	*V[μμ]	MAX(μ)	*'σ[μ]	DEP(μ)
☞ ˈmɛːʃ.tri						*			
ˈmɛʃ.tri					*W	L	*W	*W	

(88) *norma* /nɔːrm-a/

ˈnɔːr.mə	ID-QUAL	MAX	DEP	*σ[μμμ]	*'σ[μ]	*V[μμ]	MAX(μ)	*'σ[μ]	DEP(μ)
☞ ˈnɔr.mə							*		
ˈnɔːr.mə				*W		*W	L		
ˈnɔː.mə		*W				*W	*		

(89) *normal* /nɔːrm-al-0/

nɔːrˈma.l	ID-QUAL	MAX	DEP	*σ[μμμ]	*'σ[μ]	*V[μμ]	MAX(μ)	*'σ[μ]	DEP(μ)
☞ nɔrˈmaː.l						*	*		*
nɔːrˈmaː.l				*W		**W	L		*

(90) *comer* /kɔm-e-rɔ/

ku'me.ɾ	ID-QUAL	MAX	DEP	*σ[μμμ]	*'σ[μ]	*V[μμ]	MAX(μ)	'σ[μ]	DEP(μ)
ku'me.ɾ						*			*
ku'me.ɾ					*W	L		*W	L
ku'me.ɾ				*W		*			*

(91) *tia* /ti-a/

'ti.ɐ	ti.'ti.ɐ	'ti.ɐ	'ti.ɐ
IDENT-QUALITY			
MAX			
DEP			
*σ[μμμ]			
*'σ[μ]		*W	
*V[μμ]	*	L	*
MAX(μ)			
*'σ[μ]		*W	
DEP(μ)	*	L	*
*'σ[μ]		*W	
*V[low, -high, -back, μμ].V			
*σ[C ₀ V ₁ V ₂ [-high, ±ATR]]			
DEP(high)			*W
DEP(ATR)			*W

(92) *teia* /te-a/

'te.ɐ	te.'te.ɐ	'te.ɐ	'te.ɐ	'te.ɐ
IDENT-QUALITY				
MAX				
DEP				
*σ[μμμ]				
*'σ[μ]		*W		
*V[μμ]	*	L	*	*
MAX(μ)				
*'σ[μ]		*W		
DEP(μ)	*	L	*	*
*'σ[μ]		*W		
*V[low, -high, -back, μμ].V			*W	
*σ[C ₀ V ₁ V ₂ [-high, ±ATR]]				*W
DEP(high)	*	L	L	*
DEP(ATR)	*	L	L	*

(93) *ateia* /a-tɛ-a/

e'tɛ.e	ɛ'e'tɛi.e	e'tɛ.e	e'tɛ:.e	e'tɛɛ.e	e'tɛɪ.e
IDENT-QUALITY					
MAX					
DEP					
*σ[μμμ]					
*'σ[μ]		*W			
*V[μμ]	*	L	*	*	*
MAX(μ)					
*'σ[μ]		*W			
DEP(μ)	*	L	*	*	*
*'σ[μ]		*W			
*V[−low, −high, −back, μμ].V			*W		
*σ[C ₀ V ₁ V ₂ [−high, ±ATR]]				*W	*W
DEP(high)	*	L	L	L	*
DEP(ATR)	*	L	L	*	L

In Contemporary EP, the constraint ranking outlined above ensures that underlying long vowels retain their length faithfully only in stressed open syllables and in stressed closed syllables whose coda is [ʃ] or [ʒ]. In these contexts, furthermore, underlying short vowels are also realised as long vowels due to the ranking $*'σ[μ] \gg *V[μμ] \gg MAX(μ)$, which results in a surface-level neutralisation of underlying vowel quantity opposition. The only context where this opposition could potentially remain is in unstressed syllables; even there, however, $*V[μμ] \gg MAX(μ)$ enforces surface-level neutralisation of quantity opposition. In such cases, the underlying opposition is preserved through vowel quality reduction: only underlying short vowels undergo reduction in vowel quality. Thus, since **the underlying opposition in vowel quantity is either neutralised on the surface or reinterpreted as a quality opposition**, it is therefore likely that **native speakers of EP do not perceive the opposition in vowel quantity as of distinctive nature**.

5. Analysis summary

In Section 3.2, based on the phonological principle that "An underlying or phonological representation will contain all and only the unpredictable (distinctive) information for each lexical item" (Kenstowicz 1994: 60), in the alternations [á:] ≈ [ɐ] and [á:] ≈ [a], the opposition between [ɐ] and [a] is considered to be underlying, and the phonological changes [ɐ] → [á:] and

[a] → [á:] indicate that the underlying opposition between [ɐ] and [a] is neutralised in stressed syllables. On the other hand, since [ɐ] is an allophone that occurs only in unstressed syllables, its underlying form is interpreted as /a/, and by introducing the concept of mora, the phonological alternation $\overset{\mu}{/a/} \rightarrow [\acute{a}:] \approx [\text{ɐ}]$ is established. In this alternation, the quality reduction of $\overset{\mu}{/a/}$ to [ɐ] occurs, and if a similar reduction takes place in the other alternation $[\acute{a}:] \approx [a]$, it can only be a reduction in quantity, leading to the phonological alternation $\overset{\mu\mu}{/a/} \rightarrow [\acute{a}:] \approx [a]$. By extending this argument to the other relevant alternations, it is concluded that in contemporary EP, an opposition between underlying short vowels $\overset{\mu}{/e} \overset{\mu}{a} \overset{\mu}{ɔ}/$ and underlying long vowels $\overset{\mu\mu}{/e} \overset{\mu\mu}{a} \overset{\mu\mu}{ɔ}/$ has persisted. To support this interpretation, acoustic data, the length as an implementation of stress in contemporary EP and vowel coalescence in prosodic phonology are presented. In Section 4, to explain the mappings $\overset{\mu}{/i} \overset{\mu}{e} \overset{\mu}{\epsilon} \overset{\mu}{a} \overset{\mu}{ɔ} \overset{\mu}{o} \overset{\mu}{u}/ \rightarrow [\acute{i}: \acute{e}: \acute{\epsilon}: \acute{a}: \acute{o}: \acute{u}:] \approx [\acute{i} \acute{i} \acute{e} \acute{e} \acute{o} \acute{o} \acute{u}]$ and $\overset{\mu\mu}{/e} \overset{\mu\mu}{a} \overset{\mu\mu}{ɔ}/ \rightarrow [\acute{e}: \acute{a}: \acute{o}:] \approx [\epsilon \text{ a } \text{ɔ}]$, a stepwise phonological derivation from a perspective of phonological stratification is proposed, consisting of the following stages: word formation and the accompanying deletion, insertion, feature change, syllabification and stress assignment → vocalic quality modification → vocalic quantity adjustment. Correspondingly to the last two, the phonological strata are assumed to be as follows: Vocalic Quality Modification Stratum and Vocalic Quantity Adjustment Stratum. In Sections 4.1 and 4.2, the focus is narrowed down to the latter two, and the constraint hierarchies operating there are extracted. In doing so, to more accurately capture the vowel opposition in EP in terms of tongue height or aperture, the traditional feature [±ATR] is reinterpreted to represent the terminal directions of the three-stage vectorial movement leading to each height in the EP seven-vowel system.

6. Theoretical applications

In this section, we shall examine several phenomena that can be accounted for mainly by appealing to (i) the opposition between underlying short vowels $\overset{\mu}{/i} \overset{\mu}{e} \overset{\mu}{\epsilon} \overset{\mu}{a} \overset{\mu}{ɔ} \overset{\mu}{o} \overset{\mu}{u}/$ and underlying long vowels $\overset{\mu\mu}{/e} \overset{\mu\mu}{a} \overset{\mu\mu}{ɔ}/$, as posited in Section 3.2, and (ii) the phonological processes discussed in Section 4, namely vocalic quality modification and vocalic quantity adjustment, both of which are sensitive to stress and to the context in which the vowel is found and take place across the two hypothesised strata.

6.1. Lack of quality reduction of nasal vowels in unstressed syllables

Contemporary EP typically exhibits five distinct nasal monophthongs [ĩ ẽ ã õ ù] at the phonetic

level¹⁰. These vowels undergo quantity alternation [ĩ: ê: ê: ô: ú:] ≈ [ĩ ê ẽ õ ù] between stressed and unstressed syllables, but typically do not undergo the kind of vowel quality reduction characteristic of underlying short oral vowels, as seen in $p[é:]sa \approx p[i:]samos$ from the verb *pesar* "to weigh" and $m[ó:]ra \approx m[u:]ramos$ from the verb *morar* "to live". For example, the verb *pensar* "to think" shows the alternation $p[ê:]sa \approx p[ẽ:]samos$, but not $p[ê:]sa \approx *p[ĩ:]samos$; similarly, the verb *contar* "to tell, to count" exhibits the alternation $c[ô:]ta \approx c[õ:]tamos$, but not $c[ô:]ta \approx *c[ũ:]tamos$. Why, then, do these vowels not undergo quality reduction? **The fact that, in alternations between stressed and unstressed nasal vowels, only vowel quantity changes while quality remains constant is a property shared with the underlying long vowels** — /ɛ: a: ɔ:/ → [é: á: ó:] ≈ [ɛ a ɔ]. Does this, then, mean that nasal vowels are underlyingly long vowels?

When discussing nasal vowels, the initial problem lies in the phonological interpretation of the nasal vowel itself. In the case of Portuguese, there exist two competing analyses: one views the nasal vowel as *a single segment*, while the other considers it as *a phonetic realisation of a sequence of two segments*. The former is represented by works such as Lüdke (1953), but may be no longer a widely accepted view today. The latter, by contrast, branches into two theoretical traditions: (i) a functionalist-structuralist interpretation, according to which the nasal vowel is the phonetic realisation of a sequence composed of a vocalic phoneme and an archiphoneme /N/ (Barbosa 1983, 1994¹, 1994²); and (ii) an autosegmental account, in which nasal vowels derive from a sequence consisting of a vocalic segment and an autosegment [+nasal] (Mateus and Andrade 2002; Mateus et al. 2003). This study aligns with the two aforementioned *two-segment sequence analyses* and argues, on the basis of the three considerations presented below, that the phonetically realised simple nasal vowel [Ṽ] in Portuguese is phonologically /Vn/ — that is, a sequence consisting of a vocalic segment /V/ followed by a nasal consonantal segment /n/.

The first piece of evidence lies in the phonological alternation [V.n] ≈ [Ṽ.] observed within the same morpheme under different phonological environments. For instance, the alternation [i.n-] ≈ [ĩ.-] of the negative prefix *in-* in *in+ativo* [i.na'ti:vu] "inactive" vs. *in+sanável* [ĩ.sə'na:vɛɫ] "incurable" — where + marks the boundary between a derivational prefix and the base — and the alternation [ir.mə'n-] ≈ [ir.mẽ.-] of the root morpheme *irman-* in *irman+ar* [ir.mə'na:r] "to join together" vs. *irman+dade* [ir.mẽ'da:ði] "brotherhood" suggest the following: an underlying

¹⁰ In certain boundaries between clitics, or between clitics and words, one may find another nasal vowel [ã], distinct from [ẽ], as in *a antiga* [ã'ti:ya] — which will be discussed later in this chapter.

sequence /Vn/ is syllabified as {V}{nV} before another vowel /V/, whereas before a consonant-vowel sequence /CV/ it is syllabified as {Vn}{CV}, where { } indicates syllabic parsing.

In the former case {V}{nV}, the underlying /n/ is syllabified as the onset of the following syllable and is phonetically realised as segmental [n]. In the latter {Vn}{CV}, /n/ is syllabified as the coda of the preceding syllable, nasalises the preceding vowel /V/ and is then deleted at the phonetic level — {Vn}{CV} → {Ṽn}{CV} → {Ṽɳ}{CV} → {Ṽ:}{CV}. As illustrated by the alternation *cannot* ['kænə:t] ≈ *can't* ['kænt] in American English, it is a cross-linguistically common phenomenon for a vowel to become nasalised before a nasal coda. Furthermore, as seen in certain dialects of American English where *can't* is realised as [kæ̃:t], the nasal coda [n] may be deleted after nasalising the preceding vowel (cf. Kenstowicz 1994: 14). These observations suggest that the interpretation proposed above is not phonologically implausible.

The second piece of evidence concerns the phonetic realisation of the consonant /r/. This segment is realised as a tap or a flap [ɾ] when it follows an oral vowel, i.e. /VrV/ → [VɾV] — as in *caro* "expensive" /kar-o/ → ['ka:ru] — but typically as a trill [ʀ] (or a fricative [ʁ], or more conservatively as a trill [r]) when it follows another consonant such as /s/, /l/ or /r/, i.e. /VsV/, /VlV/, /VrV/ → [VʒV], [VʁV], [VrV] — as in *Israel* /israɛl/ → [iʒʀɛ'ɛʃ], *melro* "blackbird" /mɛlro/ → ['mɛʀu] and *carro* "car" /karro/ → ['ka:ʀu]. (In /VrV/, the first /r/ is assumed to be deleted following the change of the second /r/ to [ʀ], probably due to an OCP-related constraint on adjacent identical liquids.) This distribution shows that when /r/ occurs in syllable-initial position, it is realised as [ɾ] if preceded by a [−consonantal] segment and as [ʀ] if preceded by a [+consonantal] one. Now, since /r/ is also realised as [ʀ], but never as [ɾ], following a nasal vowel — e.g. [ṼrV] as in *genro* "son-in-law" ['ʒɛ:ru] — it is reasonable to assume that, phonologically, a [+consonantal] segment intervenes between the vowel and /r/. That is, **the sequence [ṼrV] should be analysed phonologically as /VnrV/.**

The third piece of evidence concerns the phonetic realisation of the consonants /b d g/. In Contemporary EP, it is generally the case that in the phonic environment [V_V], these segments assimilate the redundant feature [+continuant] of the preceding vowel and are realised as fricatives [β ð ɣ] — e.g. /VbV/, /VdV/, /VgV/ → [VβV], [VðV], [VɣV], as in *o boi* [u'βoi] "the ox", *o dia* [u'ði:ɐ] "the day", *o gato* [u'ɣa:tu] "the cat". By contrast, in the environment [Ṽ_V], these same consonants are constantly realised as stops [b d g] — e.g. [Ṽ]/bV/, [Ṽ]/dV/, [Ṽ]/gV/ → [ṼbV], [ṼdV], [ṼgV], as in *um boi* [ũ'boi] "an ox", *um dia* [ũ'di:ɐ] "a day", *um gato* [ũ'ga:tu] "a cat".

This alternation /b d g/ → [β ð ɣ] ≈ [b d g] after oral versus nasal vowels supports the analysis that [Ń_V] is phonologically /Vn_V/: since the consonantal /n/, unlike vowels, is specified as [–continuant], the following /b d g/ cannot assimilate [+continuant] from it, and their underlying [–continuant] specification is preserved in surface realisation. (For example, *longe* /lonʒe/ "far (away)" is phonetically realised as ['lõ:ʒi], and not as *['lõ:ɖʒi]; this indicates that the consonant following /n/ — in this case, /ʒ/, which is [+continuant] — does not assimilate the [–continuant] feature of /n/; it follows, then, that /b d g/ are underlyingly [–continuant] stops, since they cannot assimilate the [–continuant] feature of /n/.) From these three points, the interpretation that "the phonetically nasalised vowel [Ń] is derived from the underlying segmental sequence /Vn/" may be regarded as phonologically well grounded.

From the above discussion, two main points (94) and (95) have emerged:

(94) Nasal vowels exhibit the same type of alternation as underlying long vowels. That is, in alternations between stressed and unstressed syllables, vowel quantity changes but vowel quality remains constant — namely, [í: é: ê: ó: ú:] ≈ [ĩ ē ẽ õ ù], just as the underlying long vowels /ε: a: ɔ:/ → [é: á: ó:] ≈ [ε a ɔ].

(95) Nasal vowels are the phonetic realisation of a segmental sequence consisting of a vocalic segment followed by a nasal consonant segment, i.e., /Vn/.

The inference from (94) and (95) is as follows: **nasal vowels, at the input to the Vocalic Quality Modification Stratum and the Vocalic Quantity Adjustment Stratum, are derived long vowels of bimoraic weight** — namely, í: ē: ê: ó: ù: — just like the underlying long vowels — ε: a: ɔ:. As long vowels, they are evaluated from the same constraint hierarchy and undergo the same process as underlying long vowels with respect to both quality and quantity. Precisely because they are bimoraic, they **undergo no change in quality at the Vocalic Quality Modification Stratum** — just like underlying long vowels — and then, **at the Vocalic Quantity Adjustment Stratum, they are realised with full quantity in stressed syllables, but lose one mora in unstressed syllables**, undergoing shortening in the output — í: é: ê: ó: ú: ≈ í: ē: ê: ó: ù: → í: é: ê: ó: ù: ≈ í ē ẽ õ ù, just as é: á: ó: ≈ ε: a: ɔ: → é: á: ó: ≈ ε a ɔ. On the other hand, since they are underlyingly segmental sequences /Vn/, **the nasal vocalisation of the /Vn/ sequence — that is, the nasalisation of /V/ followed by the deletion of /n/ — must occur prior to the Vowel Quality Modification Stratum, that is, between syllabification and vowel quality modification**. Furthermore, insofar as they surface as long vowels immediately after nasal vocalisation, **the /Vn/ sequence must be bimoraic, and no mora deletion takes place**

during the process of nasal vocalisation, as shown in (96):

(96)	$\mu\mu$ /Vn/		$\mu\mu$ /V/		Input representation
	in en ɛn an ɔn on un	in en ɛn an ɔn on un	ɛ: a: ɔ:	ɛ: a: ɔ:	Syllabification
	ín éñ én án óñ óñ ún	—	é: á: ó:	—	Stress Assignment
	ĩ: ê: ê: ẽ: õ: õ: û:	ĩ: ê: ê: ẽ: õ: õ: û:	—	—	Vowel Nasalisation
	—	—	—	—	Vowel Quality Modification
	—	ĩ ẽ ẽ ẽ õ õ û	—	ɛ a ɔ	Vowel Quantity Adjustment
	ĩ: ê: ê: ẽ: õ: õ: û:	ĩ ẽ ẽ ẽ õ õ û	é: á: ó:	ɛ a ɔ	Output representation

Makino (2009: 81) derives the constraint of *rhyme maximality* in Portuguese — namely, the markedness constraint that "a rhyme (accordingly, a syllable) that exceeds three moras is inadequate" (translated from Japanese) — from the following phonotactic observations. In Portuguese, /V/ and /Vs/, which contain one [+sonorant] segment, as well as /Vr/, /Vl/, /Vi/, /Vu/, /Vis/, /Vus/, which contain two [+sonorant] segments, are all adequate as rhymes. By contrast, /Vir/, /Vur/, /Vil/, /Vul/, which contain three [+sonorant] segments, are all inadequate as rhymes. From this, it is concluded that each [+sonorant] segment in the rhyme carries one mora of weight. Since /n/ is also [+sonorant], it is predicted that it, too, carries one mora in the rhyme, just like /i/, /u/, /r/, /l/. This prediction is supported by the difference in resyllabification behaviour following stress assignment between the conjugated forms *poluis* "you pollute" and *poluem* "they/of the verb *poluir* "to pollute". In the case of *poluis* — /polu-i-s/ → {po}{lu}{is} → {po}{lu}{is} → {po}{lu}{is} → [pu'luĩ] — the two syllables {lu}{is} are resyllabified into a single syllable {'lu}{is} after stress assignment, due to the markedness constraint *'σ[μ]. In contrast, in the case of *poluem* — /polu-i-n/ → {po}{lu}{in} → {po}{lu}{in} → [pu'lu:ẽĩ] — the two syllables {lu}{in} are not resyllabified into *{'lu}{in} even after stress assignment. This is because /s/, being [−sonorant], is non-moraic, whereas /n/, being [+sonorant], is moraic, so that *{'lu}{in} would possess three moras of weight — whereas {'lu}{is} has only two moras — and would therefore be inadequate as a syllable, due to *σ[μμμ]. Accordingly, the interpretation that the /Vn/ sequence is bimoraic prior to nasal vocalisation appears to be valid.

In the vowel nasalisation, the mapping **en ɛn an ɔn on un** → **ĩ: ê: ẽ: õ: õ: û:** takes place. In particular, the mapping **ɛn an ɔn** → **ẽ: ẽ: õ:** — where **raising of the tongue height is attested** — **resembles the quality reduction of unstressed underlying short vowels /e ɛ a ɔ o/ → [i i ɐ u u]**, and can likewise be analysed as **a kind of reduction**. Because expiratory airflow is partially consumed by nasal resonance, the volume of airflow reaching the oral cavity is reduced, which results in the neutralisation of vowel quality oppositions.

Relevant to this interpretation that vowel nasalisation in Portuguese is a form of vocalic reduction, the adverb *ontem* "yesterday" may be pronounced idiolectally as ['õ:tẽĩ], with low-mid [õ], instead of ['õ:tẽĩ], with high-mid [õ] — the latter form being the phonetic representation canonically registered in dictionaries — as Barbosa (1994: 99) noted: "Não faltam em português ... variantes individuais da realização de fonemas. Muitas passam despercebidas à maioria dos utentes, como é o caso ... da realização como [õ] ou [õ] da vogal acentuada de *ontem*", which would correspond to [õ] and [õ] in IPA, respectively. This word originates from the Latin phrase *ad noctem* or *hāc nocte* and, according to Williams (1975: 94), underwent the following changes: *hac nōcte* > **a nocte* > *aõite* > *oonte* ... > *ontem*. From the etymological point of view, therefore, the *o-* in *ontem* derives from the vowel sequence *oo-* in *oonte* and is assumed to be an underlying long vowel /^{μμ}õ/: accordingly, the underlying representation of this adverb is /^{μμ}õnten/. This underlying long vowel /^{μμ}õ/ — which, in unstressed syllables, undergoes reduction through the loss of one mora while retaining its low-mid quality and is realised as a short vowel [õ] — is in parallel assumed, in syllables closed by /n/, to undergo reduction through the loss of one mora while retaining its low-mid quality, to be nasalised as ã and, with the subsequent deletion of /n/ and by compensatory lengthening, to become the derived long nasal vowel ã:, as shown in (97):

$$(97) /^{μμ}õnten/ \rightarrow \text{õ}.n.ten \rightarrow \boxed{\text{'õ}.n.ten \rightarrow \text{'ãn.tẽn}} \rightarrow \text{'ã:.tẽ:} (\rightarrow \text{'ã:.tẽĩ} \rightarrow \text{'õ:.tẽĩ})$$

There also exists, on the other hand, another phonological representation /^μõnten/ in which /^{μμ}õ/ has been shortened and, in idiolects whose lexicon stores this representation, the underlying short vowel /^μõ/ — which, in unstressed syllables, undergoes reduction through the quality modification and is realised as a reduced vowel [u] — is in parallel assumed, in syllables closed by /n/, to undergo reduction through the quality modification from low-mid õ to high-mid o, to be nasalised as õ and, with the subsequent deletion of /n/ and by compensatory lengthening, to become the derived long nasal vowel õ:, as shown in (98):

$$(98) /^μõnten/ \rightarrow \text{õ}.n.ten \rightarrow \boxed{\text{'õ}.n.ten \rightarrow \text{'õn.tẽn}} \rightarrow \text{'õ:.tẽ:} (\rightarrow \text{'õ:.tẽĩ} \rightarrow \text{'õ:.tẽĩ})$$

Under this interpretation, also in the nasalisation of /V/ in the /Vn/ sequence, the general rule of reduction applies: *underlying long vowels lose quantity, while underlying short vowels lose quality*. Additional evidence in support of this generalisation is found in the adjective *antiga* "old" /^μantig-a/ \rightarrow [ẽ'ti:ɣɐ], where the sequence /^μan-/ , with an underlying short vowel /^μa/, is realised as the qualitatively reduced nasal vowel [ẽ-]. In contrast, in the phrase *a antiga* "the

old" /a^μ#antig-a/ → [ã'ti:ɣɐ] , the initial sequence *a an-*, containing a derived long vowel a^μ, is realised as the nasal vowel [ã-], which has undergone a loss of quantity but not of quality.

Note finally that expected constraint ranking in the mapping (96) are likely to be as follows:

IDENT-QUANTITY

*[-ATR]/__N]σ >> IDENT(ATR)

*[+low]/__N]σ >> IDENT(low)

IDENT[low\ATR]

*VN >> MAX(C)

The constraint hierarchy might be as follows:

IDENT[low\ATR] >> *[+low]/__N]σ >> *[-ATR]/__N]σ >> IDENT(ATR)

*[+low]/__N]σ >> IDENT(low)

*VN >> MAX(C)

6.2. Diphthongisation of vowel segments before palatal consonants

In Contemporary EP, as Barbosa (1988: 346) points out, "hoje em dia, se encontra com frequência um *iod* entre qualquer vogal acentuada e qualquer uma das quatro consoantes palatais indicadas inicial da sílaba seguinte, e isso quer tal *iod* possa quer não possa explicar-se etimologicamente, quer tenha quer não tenha representação gráfica: palavras como *caixa*, *graxa*, *olhos*, *hoje*, *unha*, *beijo*, *velho* são pronunciadas com ou sem [i], mais ou menos nítido, e isto varia de idiolecto para idiolecto e, por vezes, no mesmo idiolecto". The terms *as quatro consoantes palatais* and *iod* refer, in IPA terms, to [ʃ ʒ ʎ ɲ] and [i], respectively. In orthographic terms, these are represented as *x*, *ch* for [ʃ]; *j*, *g* (before *i*, *e*) for [ʒ]; *lh* for [ʎ]; and *nh* for [ɲ]. The glide [i], when graphically represented, appears as *i*.

For instance, in the third-person singular forms of the verbs *relaxar* "to relax" and *taxar* "to tax" in the present indicative, the stressed vowel, spelled *a* in both cases, exhibits variation between a monophthong and a diphthong ending in [i]: *rel*[á:]*xa* ≈ *rel*[ái]*xa* and *t*[á:]*xa* ≈ *t*[ái]*xa*. What is interesting is the realisation of this radical vowel *a* in unstressed syllables. In the former case, the same type of variation between a monophthong and a diphthong — *rel*[a]*xar* ≈ *rel*[ai]*xar* —

is again observed in the unstressed syllable, just as in the stressed one. In contrast, in the latter case, only the monophthong is attested to our knowledge: *t[ɐ]xar*. What does this imply? (The following is a reinterpretation of Makino (2006¹: 25-29) within the theoretical framework adopted in this study.)

The alternation between the stressed and unstressed vowels without [i] in these verbs — namely, *rel[á:]xa ≈ rel[a]xar* and *t[á:]xa ≈ t[ɐ]xar* — coincides with the alternation of an underlying long vowel /a/ in the former, and that of an underlying short vowel /a/ in the latter, as indicated in (23). Based on this, we assume the underlying representations of these verbs to be *relaxar* /rela^μʃ-a-r0/ and *taxar* /ta^μʃ-a-r0/, respectively.

Next, let us turn to the behaviour of [i]. The variation or alternation between a monophthong — here abbreviated as V — and a diphthong — abbreviated as V_i — is limited to open syllables preceding the prepalatal and palatal consonants [ʃ ʒ ʎ n], and does not occur before [s z l n], which are adjacent to them in relation to the place of articulation feature within the consonant system. It is therefore clear that the [i] element in V_i is in some way conditioned by articulatory properties that are present in [ʃ ʒ ʎ n] but absent in [s z l n]. From a diachronic perspective, many of these consonants appear to have arisen through coalescence involving mutual assimilation between coronal consonants such as [s z l n] and the dorsal glide [j] or [ɨ]. Examples include lat. *rūssěu-* > pt. *ro*[ʃ]o; lat. *cervēsīa-* > pt. *erve*[ʒ]a; lat. *sěděa* > pt. *se*[ʒ]a; lat. *allīu-* > pt. *a*[ʎ]o, lat. *těněō* > te[n]o (cf. Williams 1961: 42, 44, 46, 50, 91). A similar process is observed in contemporary English, as in *I miss you* [A_i mis ju], which can phonetically surface as [A_i miʃu] due to the coalescence of [-s j-] into [-ʃ-] via mutual assimilation.

In Contemporary EP, the underlying segments /s z ʎ n/ are generally interpreted as simple segments with a single place of articulation. However, in the present study — taking into account both their historical development and their articulatory characteristics in the contemporary language — these palatal consonants are analysed as complex segments involving two distinct oral constrictions: a primary articulation formed by the tongue crown against the alveolar ridge — a gesture shared with /s z l n/, represented as the node **Coronal-[+anterior]** in feature geometry (cf. Kenstowicz 1994: 136-168); and a secondary articulation formed by the tongue front against the hard palate — a gesture characteristic of [j ɨ], represented as the node **Dorsal-[−back, +high]**. If phonological features are understood as neural commands issued by the brain to the speech organs in order to execute specific muscular gestures (cf. Kenstowicz 1994: 20, 138), then in the case of /ʃ ʒ ʎ n/, one may say that two distinct constriction

commands co-occur at the phonological level: a command to the tongue crown to constrict against the alveolar ridge, and a command to the front part of the tongue body to constrict against the hard palate. These co-occurring commands give rise, at the phonetic level, to articulations that range from post-alveolar to pre-palatal to palatal (for the place of articulation of [ʃ ʒ ʎ ɲ], cf. Barbosa 1994¹: 63).

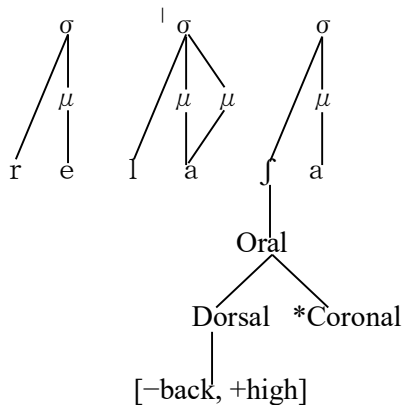
The above discussion can be summarised in the following two points:

(99) The underlying representations of the verbs *relaxar* and *taxar* are /relaʃ^μ-a-r0/ and /taʃ^μ-a-r0/, respectively.

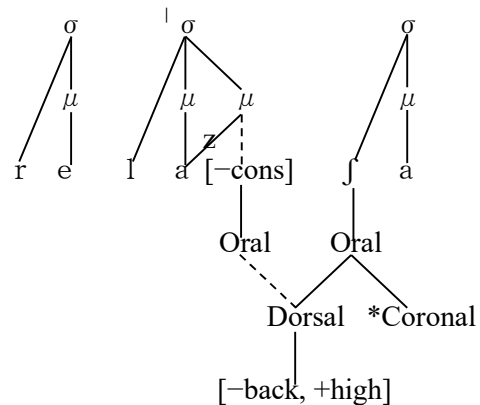
(100) The consonants /ʃ ʒ ʎ ɲ/ are analysed as complex segments involving two simultaneous oral gestures: a coronal constriction at the alveolar ridge — corresponding to the node Coronal-[+anterior] — and a dorsal constriction at the hard palate — corresponding to the node Dorsal-[−back, +high].

With the assumptions in (99) and (100), it becomes possible to account for the variation observed in *rel*[á:]*xa* ≈ *rel*[ái:]*xa* ≈ *rel*[a]*xar* ≈ *rel*[ai]*xar* and *t*[á:]*xa* ≈ *t*[ái:]*xa* ≈ *t*[ɐ]*xar*. Let us begin with the mapping /relaʃ^μ-a/ ≈ /relaʃ^μ-a-r0/ → *rel*[á:]*xa* ≈ *rel*[a]*xar*. Here, the stressed vowel [á:] is selected as the output corresponding to the input underlying long vowel /a/, evaluated as the optimal candidate under the constraint ranking in (82). The unstressed [a] is similarly derived. In contrast, in the mapping /relaʃ^μ-a/ ≈ /relaʃ^μ-a-r0/ → *rel*[ái:]*xa* ≈ *rel*[ai]*xar*, the vowel quality /a/ of the underlying long vowel /a/ is first delinked from the second mora, resulting in /a/. Then, to satisfy the vocalic constriction required by a syllable nucleus, a floating node [−consonantal, +sonorant]-Oral is introduced and linked to the delinked mora. Finally, the node Dorsal-[−back, +high] — the secondary palatal articulation inherent in /ʃ/ — spreads onto this node as in (101):

(101) [Ri'la:ʃɐ]



[Ri'laɪʃɐ]



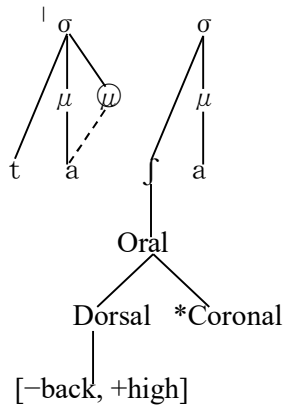
The resulting complex — $\overset{\mu\mu}{/a\underset{\cdot}{f}}/ \rightarrow \overset{\mu}{a} \overset{\mu}{f} \rightarrow \overset{\mu\mu}{a\underset{\cdot}{f}}$ — is then evaluated under the constraint ranking in (82), and its optimal phonetic realisations emerge as $[\acute{a}\underset{\cdot}{f}]$ and $[a\underset{\cdot}{f}]$ as in (102):

(102) *relaxa* /rel $\overset{\mu\mu}{a}\underset{\cdot}{f}$ -a/

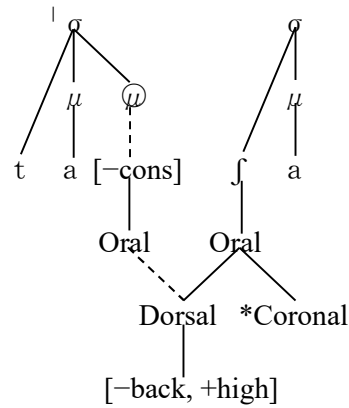
rɪˈlaː.fə	ID-QUAL	MAX	DEP	*σ[μμμ]	*'σ[μ]	*V[μμ]	MAX(μ)	*'σ[μ]	DEP(μ)
ɹɪˈlaː.fə						*			
ɹɪˈlaɪ.fə						*			
rɪˈlaɹ.fə			*W			L			
rɪˈla.fə					*W		*W		

Next, consider the mapping $/ta\underset{\cdot}{f}\text{-}a/ \rightarrow t[\acute{a}\underset{\cdot}{:}]xa$. In this case, the vowel quality $/a/$, which is associated with only one mora in the underlying representation $/a\underset{\cdot}{/}$, becomes associated with an additional mora — represented by $\textcircled{\mu}$ in (104 left) below — under stress in the Vocalic Quantity Adjustment Stratum, resulting in a long vowel $[\acute{a}\underset{\cdot}{:}]$. This long vowel is then selected as the output corresponding to the input underlying short vowel $/a\underset{\cdot}{/}$, as it is evaluated to be the optimal candidate under the constraint ranking given in (82). On the other hand, in the mapping $/ta\underset{\cdot}{f}\text{-}a/ \rightarrow t[\acute{a}\underset{\cdot}{:}]xa$, the vowel quality $/a/$ of the underlying short vowel $/a\underset{\cdot}{/}$ is not linked to the additional mora assigned under stress and is realised as a short vowel $[\acute{a}]$. In its place, an [−consonant, +sonorant]–Oral node — introduced to satisfy the syllable nucleus's requirement for vocalic constriction — is linked to that mora. Finally, the Dorsal-[−back, +high] node, corresponding to the secondary palatal articulation of $/f/$, spreads onto this Oral node, as shown in (106 right):

(103) $[\text{'ta}:\underset{\cdot}{f}\underset{\cdot}{ə}]$



$[\text{'tai}\underset{\cdot}{f}\underset{\cdot}{ə}]$



(104) *taxa* /t^μaʃ-a/

Finally, in the mapping $/t^{\mu}a\text{f-a-r0}/ \rightarrow t[v]xar$, the segment $[v]$, which is the output of the underlying short vowel $/a^{\mu}/$ in the Vocalic Quality Modification Stratum and serves as the input to the Vocalic Quantity Adjustment Stratum, occupies an unstressed syllable position. As such, it neither receives an additional mora nor associates with one, as indicated in (105):

(106) *taxar* /t^μaʃ-a-r0/

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It is worth noting that, regarding this palatal extension, one of the informants cited in Makino (2005) made the particularly intriguing observation that "after speaking for a long time and becoming tired, I tends to say *rel*[ai]*xar* instead of *rel*[a]*xar*". Phonetically, this phenomenon may be explained by a form of regressive assimilation that facilitates the articulatory transition from $\overset{\mu}{/a/}$ or $\overset{\mu}{/a/}$ to $/f/$.

6.3. Phonetic realisation of vowel coalescence in prosodic phonology

It was already noted in Section 3.2 that the [a] in the imperative sentence *Fica aqui* "Stay here" → *Fic*[a]*qui*, where the locative adverb *aqui* follows the second person singular imperative form *fica* of the verb *ficar* "stay", can be interpreted as the phonetic realisation of a derived long vowel a at the phrasal level. A similar phenomenon is also observed within the domain of the prosodic word:

(107) *Dava-a*. /dav-a#a/ → ['da:va] "He/she/you used to give it."

(108) *Dava-o*. /dav-a#o/ → ['da:vɔ] "He/she/you used to give it."

Example sentence (107) is of the same type as *Fic*[a]*qui*, and appears to derive as follows: *Dava*^μ*-a* → *Dava*^μ*-a*^μ → *Dav*^μ*'-a*. In this process, the vowel quality of the first segment *a* is deleted probably due to the OCP, the remaining mora is relinked to the following *a*, and a derived long vowel a is formed, which is realised as [a] at the surface level.

Example (108) is slightly more complex: the [+ATR] feature of the vowel /o/ is delinked from the Dorsal node in the feature geometry, allowing the spreading of the [−ATR] feature of the preceding /a/¹¹. Then, /a/ itself is delinked from its mora at the root node level, and a segment

¹¹ The personal pronoun *o* [u] is a clitic lexically specified as unstressed, and since no alternation with a stressed vowel exists, its underlying values for [high] and [ATR] are indeterminate. Theoretically, it could be /u/, /o/ or /ɔ/. However, based on orthographic evidence that an underlying /u/ is consistently written as *u* even in word-final position, as in *venús-io* ≈ *venus-to* ≈ *Vénus* and *onusto* ≈ *ónus*, whereas the nominal/adjectival thematic marker (índice temático) *-o* is written as *o* and are unlikely to be derived from an underlying /u/ — for example, the underlying forms of *vau* "ford", *véu* "veil", and *feio* "ugly", *fio* "string" may be analysed as /va/ + *-o*, /ve/ + *-o* and /fe/ + *-o*, /fi/ + *-o*, respectively; when the vowel preceding *-o* bears stress, these forms undergo two different types of syllabification; the former group surfaces as monosyllabic words [váu}, [véu}, whereas the latter as disyllabic words [fêi}{u}, [fi:}{u}, where { } indicates a syllable; this difference in syllabification can be straightforwardly explained if the underlying form of *-o* is assumed to be /o/ or /ɔ/. If *-o* is analysed as /o/, the underlying representations of these forms are /va-o vɛ-o/ and /fe-o fi-o/, respectively; in other words, when the vowel preceding *-o* /o/ is lower than /o/, as in /a ɛ/, monosyllabification occurs; when it is equally or more close, as in /e i/, disyllabification occurs; the same type of explanation holds if /ɔ/ is assumed as the underlying form; if /u/ is assumed, however, the difference between monosyllabification and disyllabification can no longer be explained in terms of the relative aperture between the preceding

with altered feature values is relinked to the vacated mora, forming the derived long vowel $\overset{\mu\mu}{\mathfrak{a}}$, as shown in (109), where the features retained from the two original segments in this coalescence are shaded. This vowel is subsequently shortened by one mora under constraint hierarchy (82), surfacing as the optimal candidate [ɔ]:

(109)

	$\overset{\mu}{/a/}$	$\overset{\mu}{/o/}$		$\overset{\mu\mu}{\mathfrak{a}}$
low	+	–	⇒	–
high	–	–		–
ATR	–	+		–
back	+	+		+
round	–	+		+

The derivation of long $\overset{\mu\mu}{a}$ within a prosodic word is also observed in forms such as the contraction of the preposition *a* [ɐ] + the definite article *a* [ɐ] → *à* [a], and the contraction of the preposition *a* [ɐ] + the demonstrative *aquilo* [ɐ'ki:lu] → *àquilo* [a'ki:lu]. The fact that *a* and *à* or *aquilo* and *àquilo*, are phonetically distinguishable in EP is due to the survival of the vowel quantity opposition. In BP, by contrast, the long vowels have been eliminated from the vowel system, and the opposition of vowel quantity has collapsed — for example, *pr[i]gar* "to nail", with short $\overset{\mu}{/ɛ/}$, and *pr[ɛ]gar* "to preach", with long $\overset{\mu\mu}{/ɛ/}$, which are distinguished by underlying vowel quantity in EP, are homophones *pr[ɛ]gar* with $/ɛ/$ in BP. As a result, *a* and *à* or *aquilo* and *àquilo*, are not phonetically distinguishable. The inability to distinguish in speech the contracted form *à* (from *a* + *a*) from either the simple preposition *a* or the simple article *a* constitutes a significant communicative problem. It may well be one of the reasons why, in contemporary spoken BP, alternative prepositions such as *para* or *em* are often used in contexts where *a* would appear in writing.

6.4. The distinctive opposition [ɛ:] ≈ [á:] in *-amos* and *-ámos* in the first-person plural of the present and preterite indicatives

In stressed CV syllables preceding [+nasal] consonants [m n ɲ], $/a/$ is realised as [ɛ:], reflecting a change from [+low] to [–low] — e.g. *cama* "bed" /kam-a/ → ['kɛ:mɐ]; *ano* "year" /an-o/ → ['ɛ:nu]; *banho* "bath" /baɲ-o/ → ['bɛ:ɲu]. This raising may be regarded as a type of *réduction*,

vowels $/a \ \varepsilon \ e \ i/$ and $-o \ /u/$ - it seems thus relatively unlikely that the underlying form of the pronoun spelled as *o* is $/u/$.

possibly resulting from airflow redistribution: nasal resonance in vowels, triggered by the spreading of the [+nasal] feature from the following nasal consonant, diverts part of the pulmonic airstream into the nasal cavity, thereby reducing the amount of airflow available for oral articulation and rendering the [+low] region inaccessible, which requires more airflow.

This phonological context also allows, nevertheless, for occurrences of the [+low] vowel [á:], though in highly restricted environments. These include the noun *g[á:]nho* "gain, profit" and the rizotonic forms of the verb *g[a]nhar* "to gain" with the same root – namely, *g[á:]nho*, *g[á:]nhas*, *g[á:]nha*, *g[á:]nham*, *g[á:]nhe*, *g[á:]nhes*, *g[á:]nhem* — as well as the first person plural of the preterite indicative in *-ar* regular verbs, encoded by the ending *-ámos* [-á:muʃ]. This final form stands in systematic opposition to the first person plural of present indicative ending *-amos* [-é:muʃ], as exemplified by *fal[á:]mos* "we spoke" vs. *fal[é:]mos* "we speak", both from the verb *falar* "to speak".

Accordingly, even in the context preceding syllable-initial [+nasal] consonants, the mid vowel [é:] and the low vowel [á:] stand in a distinctive opposition. The vowel *a* in the root *ganh-* is historically a long vowel of bimoraic weight, as noted by Teyssier (1981: 42): "Temos, com efeito, *ga-anha* > *ganha* (verbo) e *ga-anho* > *ganho* (substantivo), nos quais o *a*, resultante da contracção, conservou até hoje no português europeu um timbre aberto [a], apesar da presença da consoante nasal seguinte, que, nas palavras que contêm um *a* singelo etimológico, sempre fechou esta vogal em [ä]", which is [ɐ] or [ə] in IPA. And given that it does not undergo quality reduction even in unstressed syllables, as in *g[a]nhar*, it is reasonable to assume that it still retains its bimoraic status in contemporary usage, reflecting the alternation schema $\overset{\mu\mu}{/a/} \rightarrow [á:] \approx [a]$ discussed in (23). **This vowel quantity of $\overset{\mu\mu}{/a/}$ may thus be interpreted as blocking the raising process that would otherwise occur before nasal consonants.**

As previously noted in Makino (2007¹: 3), "in the preterite indicative, each combination of 'indicative-nonpast-finished' and a specific 'person-number' set appears morphologically as an *amalgam* — e.g. *fale-i*, *fala-ste*, *falo-u*, *falá-mos* (← *fala-* + *-^μmos*), *fala-stes*, *fala-ram*" (translated from Japanese). On the notion of *amalgam* — *amalgame* in French — Martinet (1991: 102) writes: "deux signifiés qui coexistent dans un énoncé enchevêtrant leurs signifiants de telle façon qu'on ne saurait analyser le résultat en segments successifs". The first person plural ending of the preterite indicative *-^μmos* — that is, *-^μmos* in the citation above — is thus a morphological amalgam of [–non-assertive, –VP∈past, –finished, –assumptive] with [–non-participant, –addressee, +plural]. **It begins not with a segmental element but with a floating mora $\overset{\mu}{/ /}$** , much like "the Hebrew definite prefix *ha*", which "has the phonological form *haX*,

where X is an empty position that prefers to be occupied by the following consonant but when this is impossible then by the preceding vowel" (Kenstowicz 1994: 44-45). In contrast, the **-mos** of the present indicative encodes only the person-number features [–non-participant, –addressee, +plural] and therefore lacks such a floating mora $^{\mu}$ /. If one assumes then that the thematic vowel a of the verbal stem — for example, the vowel a of the stem *fala-* of the verb *falar* — becomes linked to this initial mora, yielding a bimoraic structure as in (110), the form *fal[á:]mos* should be analysed on a par with */gaŋ-o/ → g[á:]nho*. By contrast, in (111), such mora linking does not occur due to the absence of a floating mora:

(110) preterite indicative: $fala- + -mos \rightarrow fala-^{\mu}mos \rightarrow fala^{\mu}-mos \rightarrow fal[á:]mos$

(111) present indicative: $fala- + -mos \rightarrow fala^{\mu}-mos \rightarrow fal[é:]mos$

As discussed in section 6.1, the tautosyllabic vowel-nasal sequence Vn undergoes nasalisation of V and deletion of n with the compensatory lengthening of $\tilde{V}$ — namely, in $en \rightarrow \tilde{e}n$, $on \rightarrow \tilde{o}n$, $un \rightarrow \tilde{u}n$ — prior to the Vowel Quality Modification Stratum. The tongue-height raising of $an \rightarrow \tilde{e}n$ during this process can be attributed to the influence of [+nasal] from the following nasal segment n . Similarly, as argued just above, the raising of short $\acute{a}$ to $\acute{e}$ in stressed CV syllables preceding $[m, n, \eta]$ is likewise attributable to the [+nasal] feature from these consonants. Given that both the raising in a [+nasal]. $\rightarrow \tilde{e}$: and that in $\acute{a}$. [+nasal] $\rightarrow \acute{e}$. [+nasal] — where $.$ denotes a syllable boundary — are to be triggered by the same feature [+nasal], it seems unlikely that they should occur at different strata. If we therefore assume that the latter process also precedes the Vowel Quality Modification Stratum, then the short vowel $/a/$ and the long vowel $/a^{\mu}/$ that occur in stressed CV syllables before [+nasal] consonants would be realised as $\acute{e}$ and $\acute{a}$:, respectively, at the input to that stratum. At this stage, they are evaluated as optimal intermediate outputs by the set of constraints (64) operative in this stratum, and passed on as inputs to the subsequent Vowel Quantity Adjustment Stratum. As shown in the tableaux in (112) and (113), they are then output as $\acute{e}$: and $\acute{a}$:, respectively:

(112) *falamos* $/fal-a^{\mu}-mos/$

$f\acute{e}'l\acute{e}.mu\int$	ID-QUAL	MAX	DEP	* $\sigma[\mu\mu\mu]$	* $\sigma[\mu]$	* $V[\mu\mu]$	MAX(μ)	* $\sigma[\mu]$	DEP(μ)
$f\acute{e}'l\acute{e}.mu\int$						*			*
$f\acute{e}'l\acute{e}.mu\int$					*W	L		*W	L
$f\acute{e}'la:.mu\int$	*W					*			*

(113) *falámos* /fal-a-^μmos/ → falá^{μμ}mos

fe'la:.muɸ	ID-QUAL	MAX	DEP	*σ[μμμ]	*'σ[μ]	*V[μμ]	MAX(μ)	*'σ[μ]	DEP(μ)
☞ fe'la:.muɸ						*			
fe'la.muɸ					*W	L	*W	*W	
fe'le:.muɸ	*W					*			

In the preterite indicative of regular *-ar* verbs, the endings *-i*, *-ste*, *-u*, *-stes* and *-ram* are all morphological amalgams encoding both the TAM feature bundle [−non-assertive, −VP∈past, −finished, −assumptive] and the person-number values [±non-participant, ±addressee, ±plural]. In contrast, the present indicative endings — *-o*, *-s*, *-Ø*, *-is*, and *-m* — encode only the person-number values [±non-participant, ±addressee, ±plural] and lack any temporal-aspectual specification. It is therefore highly implausible, from a systemic point of view, to assume that only the first person plural of the preterite should share its ending *-mos* with the present indicative. Such an assumption would disrupt the overall morphological and semantic parallelism of the system and thus seems theoretically unnatural.

Accordingly, **the surface contrast between [é:] and [á:] in stressed CV syllables preceding nasal consonants [m n ɲ]** is best interpreted as **the phonetic realisation of an derived opposition between a^μ and a^{μμ}**. On this analysis, thus, stressed CV syllables preceding [+nasal] consonants [m n ɲ] constitute the only type of stressed syllable in which the opposition between the short vowel a^μ and the long vowel a^{μμ} does not undergo neutralisation — precisely because of the quality weakening affecting the short vowel.

On the other hand, the situation is different for regular verbs of the *-er* and *-ir* conjugations. The theme vowels *e* and *i* of these verbs are the underlying short vowels /e^μ/ and /i^μ/, respectively. When these vowels are linked to the floating mora / / of the preterite indicative first person plural ending -^μmos, they form derived long vowels e^{μμ} and i^{μμ}. In stressed CV syllables, however, the surface realisation of these derived long vowels is identical to that of their short counterparts — both are realised phonetically as [é:] and [í:], respectively, and thus the opposition is neutralised at the phonetic level. This interpretation is based on the following inference: as shown in (23), the underlying long vowels /ε^{μμ}/, /a^{μμ}/, and /ɔ^{μμ}/ are mapped to [é:], [á:], and [ó:], respectively, in stressed CV syllables and, if systemic consistency and parallelism are to be maintained, then the derived long vowels e^{μμ} and i^{μμ} must likewise be mapped to [é:] and [í:], which coincide with the mapping of the underlying short vowels /e^μ/ and /i^μ/ in the same prosodic environment, also illustrated in (23). This is illustrated in (114) and (115) by comparing the first

person plural form *comemos* in the preterite indicative with the superficially identical form *comemos* in the present indicative of the verb *comer* "to eat".

(114) preterite indicative: $\overset{\mu}{come-} + \overset{\mu}{-mos} \rightarrow \overset{\mu}{come-} \overset{\mu}{mos} \rightarrow \overset{\mu\mu}{come-mos} \rightarrow com[\acute{e}:]mos$

(115) present indicative: $\overset{\mu}{come-} + -mos \rightarrow \overset{\mu}{come-mos} \rightarrow com[\acute{e}:]mos$

6.5. Cases requiring further consideration

(i) Semantic transparency of morphemes and their prosodic structure

In some cases, semantic transparency of morphemes and their prosodic structure appear to be related to whether a vowel undergoes reduction. In *televisão* "television" and *telefone* "telephone", the prefix *tele-* /tele-/ meaning "remote" is realised with reduced vowels as [tɨli-]. By contrast, in *telemóvel* "mobile phone", the same string /tele-/, originated as a clipped form of *telefone*, is realised with non-reduced vowels [telɛ-]. While this semantic difference may play a role in the preservation of full vowel quality, it is also possible that, in *telemóvel*, the element *tele-* constitutes a prosodic word on its own, as a semantically transparent lexical item — unlike the *tele-* in *televisão* or *telefone*. In that case, the primary stress of the prosodic word would fall on [ˈtɛ-]. The remaining issue concerns the final unstressed vowel [-ɛ], but this may be treated analogously to the final unstressed [-ɔ] in *euro* "euro (currency)", when pronounced as [ˈɛʁɔ].

In Makino (2010), based on the observation that native speakers perceive the stress position in *papel* and *doutor* to be unmarked, just like that in *futuro*, *camada* and *solene*, a theme vowel consisting solely of one mora without segmental quality, representable by -0 — treated as having the same prosodic status as the theme vowels -o, -a, -e in *futur-o*, *camad-a*, *solen-e*, which are referred to in Portuguese linguistics as *índices temáticos* "stem markers" — was postulated, and *papel* and *doutor* were thereby allowed to be analysed as *papel-0* and *doutor-0*. On this basis, all of these words were assumed to receive their primary stress through binary foot parsing as in fu(ˈtu.r-0), ca(ˈma.d-a), so(ˈle.n-e), pa(ˈpe.l-0), dou(ˈto.r-0). In contrast, words such as *câmar-a* and *túnel-0* were assumed to contain the morphemes *camar-* and *tunel-*, which behave as lexical exceptions in stress assignment by requiring that their final syllable should be the starting point of binary foot parsing when adjacent to an *índice temático* -o, -a, -e or -0. Consequently, stress is assigned through metrification as (ˈcâ.ma)r-a and (ˈtú.ne)l-0.

Accordingly, since *tele-* and *euro* each constitute a single morpheme — with neither **tel-* nor **eur-* bearing autonomous meaning — the prosodic word structures of *tele-* and *euro* are to be

understood like ('câ.ma)r-a and ('tú.ne)l-0, respectively, and are thus metrified as ('te.le)-0 and ('eu.ro)-0. The corresponding underlying representations are /tɛ̃ː/ and /eurõː/, in which the radical final vowels /ɛ/ and /ɔ/ become associated with the floating mora of the *índice temático* -0 to form derived long vowels ɛ̃ː and ɔ̃ː. The surface realisations [ɛ̃] and [ɔ̃] of the final vowels of *tele-* and *euro*, respectively, thus appear to reflect these derived long vowels.

(ii) The absence of vowel reduction in unstressed syllables of derived words with negative connotations

Makino (2007²) investigated and classified words that, although etymologically assumed to have an underlying short vowel, do not exhibit its reduction in unstressed syllables. That analysis was based on the phonetic transcriptions from multiple dictionaries. The procedure was as follows:

(a) The phonetic transcriptions of all the headwords in Academia das Ciências de Lisboa (2001), *Dicionário da língua portuguesa contemporâneo*, Lisboa: Verbo (2 volumes, 69,426 entries), and Teixeira, G. (Ed.) (2004), *Grande dicionário da língua portuguesa*, Porto: Porto Editora (approximately 100,000 entries) were examined, and all the words with unstressed vowels that do not exhibit reduction were extracted (as of the time of writing, the first volume of Academia das Ciências de Lisboa (2001) and headwords starting with A-F in Teixeira, G. (Ed.) (2004) had been completed).

(b) Porto Editora (2001), *Dicionários Editora da Língua Portuguesa*. Porto: Porto Editora; Larousse (1996), *Larousse de poche: Dictionnaire français-portugais / português-français*. Paris: Larousse; Porto Editora (1982), *Dicionário da Língua Portuguesa* (5th ed.). Porto: Porto Editora; Fonseca, F. V. P. da (1957), *Dictionnaire français-portugais / português-français Apollo*. Paris: Larousse were examined to verify how the headwords in (a) were phonetically transcribed and to check for any variation in vowel reduction.

(c) Based on the data obtained in (a) and (b), common features of forms with vowels that, although etymologically assumed to be monomoraic, do not undergo reduction in unstressed syllables were extracted.

The descriptions in (ii)-(iv) of this subsection 6.5 are based on the reinterpretation of the findings obtained in that study. Here in (ii), we shall discuss derived words with negative connotations.

First, in nouns and adjectives derived from a nominal radical, there are many words that retain a non-reduced vowel losing stress due to derivation and are simultaneously found to carry negative connotations, especially pejorative or contemptuous connotations. What is noteworthy in those examples is that, whereas derived forms with pejorative connotations do not exhibit vowel reduction, derived forms without such connotations may exhibit vowel reduction, even though they are derived from the same radical. For example, the absence or presence of reduction of the radical vowel /ɛ/ in *ch[ɛ]f-ão* "authoritarian head, leader, boss, manager" and *ch[i]f-ia* "headship, leadership", both derived from *ch[ɛ:]f-e* "head, leader, boss, manager" demonstrates a semantic contrast, as does the difference between *c[ɛ]gu-eta* "person who sees badly", *c[ɛ]g-ada* "group of blind people" and *c[i]g-ar* "to blind", *c[i]gu-eira* "blindness", all derived from *c[ɛ:]g-o* "blind person". Other interesting examples are *a-b[o]b-ar*, *a-pal[ɛ]rm-ar* and *a-pat[ɛ]t-ar* — all meaning "to make silly, to make foolish" — derived from *b[ó:]bo* "fool", *pal[ɛ:]rma* "fool, silly" and *pat[ɛ:]ta* "fool, idiot", respectively. These are treated as synonyms in Academia das Ciências de Lisboa (2001), Teixeira, G. (Ed.) (2004), and Houaiss, A. (2001), and all of them exhibit non-reduction of the radical vowel, which is unlikely to be a mere coincidence. Second, in the derivation of such nouns and adjectives, the frequent use of suffixes that inherently carry pejorative connotations is observed. Typical examples include *-aça*, *-ice*, *-alho*, but other suffixes such as *-ão* and *-eta* may also often carry pejorative or contemptuous connotations, while *-ada*, when used to indicate collectivity, is a formative morpheme that frequently carries a pejorative connotation. The use of these suffixes makes explicit the pejorative nature of the words in which the non-reduction of the radical vowel is in question. Third, among words with negative connotations, there are those that show variation in vowel reduction, such as *bod[ɛ:]g-a* "tavern" → *bod[ɛ]gu-ic-e* ~ *bod[i]gu-ic-e* "rubbish".

The above points suggests that there may be some correlation between the presence or absence of pejorative connotations and the reduction or non-reduction of the radical vowel. But why is non-reduction of unstressed vowels observed in words with such meanings? It is possible that mora addition, which represents emotional emphasis, is involved. For instance, in French, under *accent d'insistance*, a mora is added to the *p* of *impossible* [ãpɔsibl], for example, resulting in *imppossible* [ãp̃ɔsibl], as in *Cet enfant est imppossible* "This child is absolutely uncontrollable". Similarly, in EP, emotional emphasis often results in the lengthening of the radical vowel, as in *golo* "goal" → *goooooolo* "gooooooal!"

In this regard, a relevant passage was found on *Ciberdúvidas*. It was published as an *aditamento* in response to a question regarding the use of the subject pronoun *você*, and reads as follows:

"Tenho encontrado, aqui em Portugal, essa expressão utilizada com uma carga negativa. Oiço por vezes uma pessoa, como que querendo desvalorizar a outra, exclamar, com trejeitos de desprezo e carregando e prolongando excessivamente no «cê»: «você...» ...Rui Ramos 03/02/00"¹². What is clear from this is that in EP, the lengthening of the final vowel *ê* in *você* is clearly associated with contempt or derision.

If the lengthening of the root vowel – that is, the addition of a mora to the root vowel — which expresses emotional emphasis is a process that can occur not only at the phrasal level as shown above, but also at the word formation level, then the presence or absence of reduction, as in *bod[ɛ:]g-a* → *bod[i]gu-ic-e* ~ *bod[ɛ]gu-ic-e*, could possibly arise through the derivational steps as shown in (116):

(116)

/bod ^μ ɛg-/	/bod ^μ ɛg-/	Radical
—	/bod ^{μ+μ} ɛg-/	Mora addition indicating emotional emphasis
[bod ^μ ɛg]is-e	[bod ^μ ɛg]is-e	Suffix and thematic vowel addition
bo.dɛ'.gi.se	bo.dɛ'.gi.se	Syllabification
bo.dɛ' ^μ gi.se	bo.dɛ' ^μ gi.se	Stress assignment
bu.dɪ' ^μ gi.sɪ	bu.dɛ' ^μ gi.sɪ	Vowel quality modification
bu.dɪ'gi:.sɪ	bu.dɛ'gi:.sɪ	Vowel quantity adjustment
[bu.dɪ'gi:.sɪ]	[bu.dɛ'gi:.sɪ]	Output

This additional mora to the radical vowel — if this hypothesis is correct — might, from the perspective of word formation, represent a type of emphatic *infix* composed solely of one mora of quantity, lacking in quality — that is, a *prosodic infix of emphasis*.

As mentioned above, even though pejorative connotations are frequently associated with the suffixes used in the derivation of nouns and adjectives with non-reduced unstressed vowels, this does not mean that the addition of these suffixes automatically triggers the non-reduction of the root vowel. For example, the alternation *c[á:]ra* "face" ≈ *c[ɐ]r-aça* "broad and full face (with disproportionate features)", *fanch[ó:]na* "woman with a mannish appearance and traits generally associated with men" ≈ *fanch[u]n-aça* "idem, more emphatic", *p[ɛ:]rna* "leg" ≈ *p[i]rn-aça* "chunky leg"; *[á:]sno* "fool, ass" ≈ *[ɐ]sn-ice* "foolishness", *b[ɛ:]sta* "stupid" ≈ *b[i]st-ice*

¹² Rui Ramos, who wrote this, is a journalist from Luanda, the capital of Angola. Although this aditamento has now been removed from *Ciberdúvidas*, it is still available for viewing at: <https://web.archive.org/web/20030909182445/http://ciberduvidas.sapo.pt/php/resposta.php?id=5100>.

"stupidity", *idi[ó:]ta* "idiot" $\approx$ *idi[u]t-ice* "idiocy"; *fr[á:]de* "monk" $\approx$ *fr[ɐ]d-alh-ada* "monks, monkdom", etc. Furthermore, the non-reduction of unstressed root vowels can also occur when a suffix that does not carry pejorative connotations is added — for instance, *b[ó:]bo* $\approx$ *b[o]b-agem* "nonsense"; *b[ó:]rga* "merrymaking" $\approx$ *b[ɔ]rgu-ista* "merrymaker", *b[ó:]rla* "free entrance" $\approx$ *b[ɔ]rl-ista* "person who attends performances without paying intentionally"; *br[é:]jo* "uncultivated land that only produces heather" $\approx$ *br[ɛ]j-eiro* "villain, scoundrel, knave"; *b[ó:]bo* $\approx$ *a-b[o]b-ar*, *pal[é:]rm-a* $\approx$ *a-pal[ɛ]rm-ar*, *pat[é:]ta* $\approx$ *a-pat[ɛ]t-ar*, etc. These facts suggest that **the above process of mora addition to the radical vowel itself is independent of the addition of such pejorative suffixes**. Moreover, many of these words show variation in vowel reduction across different dictionaries, and not all derived words with negative connotations exhibit non-reduction of the unstressed root vowel. This suggests that **the process of mora addition to the root vowel indicating emotional emphasis might resemble a type of affixation to free morphemes** (here, morphemes that can function as standalone words without the need for the addition of other affixes, except for the addition of the thematic vowel, are referred to as free morphemes).

This is because, first, the variation in the reduction of the radical vowel implies that morphemes that can undergo mora addition to their radical vowel can also occur without it, just as free morphemes, excluding the root vowel, do not require the addition of other affixes to function as a word. Second, the fact that not all derived words with negative connotations exhibit non-reduction of the unstressed root vowel suggests the existence of morphemes that cannot undergo mora addition due to emotional emphasis, paralleling the fact that not all affixes can be added to all free morphemes.

(iii) Derived nouns and adjectives formed by suffixation directly to the verbal radical without an intervening thematic vowel.

In Portuguese, one common method of deriving nouns and adjectives from verbs involves the addition of suffixes such as *-ção* (underlyingly presumed as *-tion*), *-dor*, *-tiv-o*, and *-tóri-o* to the verbal radical. In this derivational process, a thematic (or theme) vowel is typically inserted between the verbal radical and the suffix (e.g. [*compar-a*]*r* $\rightarrow$ [*compar-a*]*ção*; [*bat-e*]*r* $\rightarrow$ [*bat-e*]*dor*; [*part-i*]*r* $\rightarrow$ [*part-i*]*tiv-o*; [*fal-a*]*r* $\rightarrow$ [*fal-a*]*tóri-o* — square brackets [] indicate the verbal stem, and underlining denotes the verbal radical). In these derived nouns and adjectives, word stress falls on the suffix — *-ção*, *-dor*, *-tiv-*, *-tóri-* — rendering the radical vowel

unstressed. When this vowel is *a*, it undergoes quality reduction — e.g. [comp[ɐ]r-a]çãõ; [b[ɐ]t-e]dor; [p[ɐ]rt-i]tiv-o; [f[ɐ]l-a]tóri-o.

In contrast with this type of word formation, when such suffixes are attached directly to the verbal radical without an intervening thematic vowel, they take the forms *-ãõ* (underlyingly presumed as *-ion*), *-or*, *-iv-o*, *-óri-o*, and the radical vowel *a* often does not undergo reduction as in (117):

$$(117) [dil[\mathbf{v}]t-\mathbf{a}]r \rightarrow [dil[\mathbf{v}]t-\mathbf{a}]çãõ, [dil[\mathbf{v}]t-\mathbf{a}]dor \Leftrightarrow [dil[\mathbf{a}]ç]ãõ, [dil[\mathbf{a}]t]óri-o$$

$$[per-su[\mathbf{v}]d-\mathbf{i}]r \rightarrow [per-su[\mathbf{v}]d-\mathbf{i}]çãõ \Leftrightarrow [per-su[\mathbf{a}]s]ãõ, [per-su[\mathbf{a}]s]iv-o, [per-su[\mathbf{a}]s]óri-o$$

$$[in-f[\mathbf{i}]r-\mathbf{i}]r \rightarrow [i-l[\mathbf{a}]ç]ãõ, [i-l[\mathbf{a}]t]iv-o$$

The last example involves a suppletive alternation of the verbal root between *fer-* and *lat-*, but is included here because the lack of vowel reduction appears to correlate with the morphological structure of the word. Barbosa (1988: 353) makes an observation of particular interest in this regard, noting that "... o seu emprego (= o emprego de [a]: author's note) se alarga a palavras onde vem alternar com [ə] normal (= IPA [ɐ]: author's note), como... *dissuasão, persuasão...*". This suggests that **the non-reduction of the radical vowel *a* in the morphological context [verbal radical + non-thematic vowel] + [suffix *-ãõ, -iv-, -or, -óri-*] is a phenomenon currently in progress from a diachronic perspective.**

It remains unclear whether this phenomenon results from analogy with some existing paradigm. From a morphophonological standpoint, however, it is possible to account for the data by assuming that the radical vowel *-a*, even when not reduced, is underlyingly a short vowel /^μa/. On this basis, in surface forms that appear to lack a stem vowel, one may posit, as shown in (118), a stem vowel *-0* with no segmental quality but with a vocalic quantity of one mora — just like the other stem vowels *-a, -e, -i* — thereby rendering the phenomenon theoretically tractable:

$$(118) [dil\overset{\mu}{a}t-\overset{\mu}{a}]r \rightarrow [dil\overset{\mu}{a}t-\overset{\mu}{a}]çãõ$$

$$[dil\overset{\mu}{a}t-\overset{\mu}{a}]r \rightarrow [dila\overset{\mu}{ç}-\overset{\mu}{0}]ãõ$$

Then, in forms with the stem vowel *-0*, we may assume that the radical vowel /*a*/ is linked to the floating mora that constitutes *-0*, yielding a derived long vowel ^{μμ}a. This ^{μμ}a, like an underlying long /^{μμ}a/, resists quality reduction even in unstressed syllables as shown in (119):

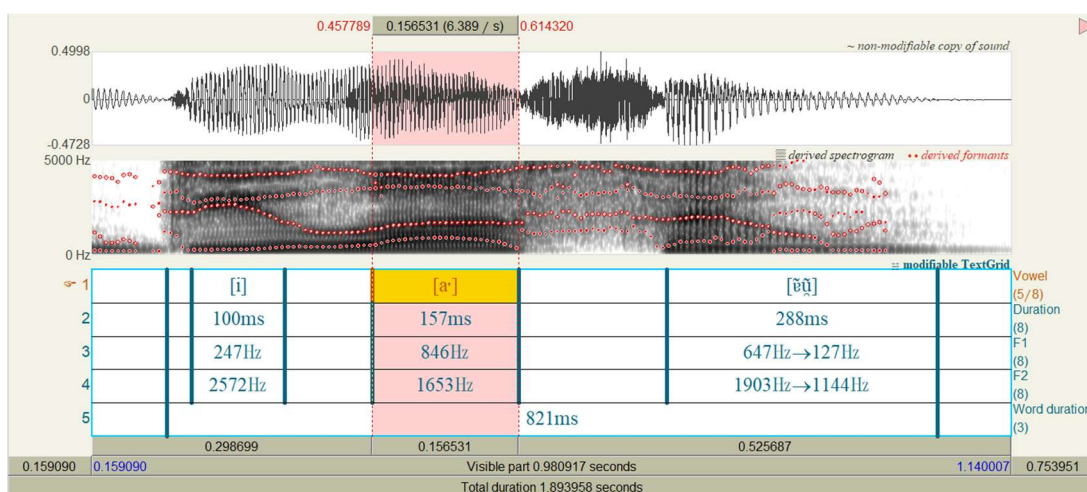
$$(119) dila\overset{\mu}{ç}-\overset{\mu}{a}õ \rightarrow dila\overset{\mu\mu}{a}\overset{\mu}{ç}-\overset{\mu\mu}{a}õ \rightarrow dil[\mathbf{a}]\overset{\mu\mu}{ç}-\overset{\mu\mu}{a}õ$$

The following are spectrographic analysis images of the audio files available for hearing on

Infopédia Dicionários Porto Editora, referenced in footnote 12. Figure (120) corresponds to *dilação*, and Figure (121) to *dilatação*¹³. The analysis was conducted using Praat (version 6.4.35). As far as we can tell, it is the same native female speaker as (14), (15), (18) and (19). The overall word durations are 821ms for *dil[aː]ção* and 923ms for *dil[ɐ]tação*, probably due to the overall syllable counts. The acoustic measurements of the presumed morphologically same radical vowel *-a-* in *dilação* and in *dilatação* are given in (122) and (123), respectively.

In (120), the radical vowel [aː], which is assumed to have two moras, in *dil[aː]ção* exhibits double the duration (157ms) and a lower quality (F1 = 846Hz) compared to the presumed morphologically identical vowel [ɐ], which is assumed to have only one mora, in *dil[ɐ]tação* (79ms, F1 = 413Hz) in (121). These examples also, like *p[aː]daria* in (14) and *C[ɐ]tarina* in (15), are likely to provide phonetic evidence that the qualitatively non-reduced unstressed vowel [a] may exhibit longer duration than the qualitatively reduced unstressed vowel [ɐ] in Contemporary EP, reflecting the difference in underlying quantity.

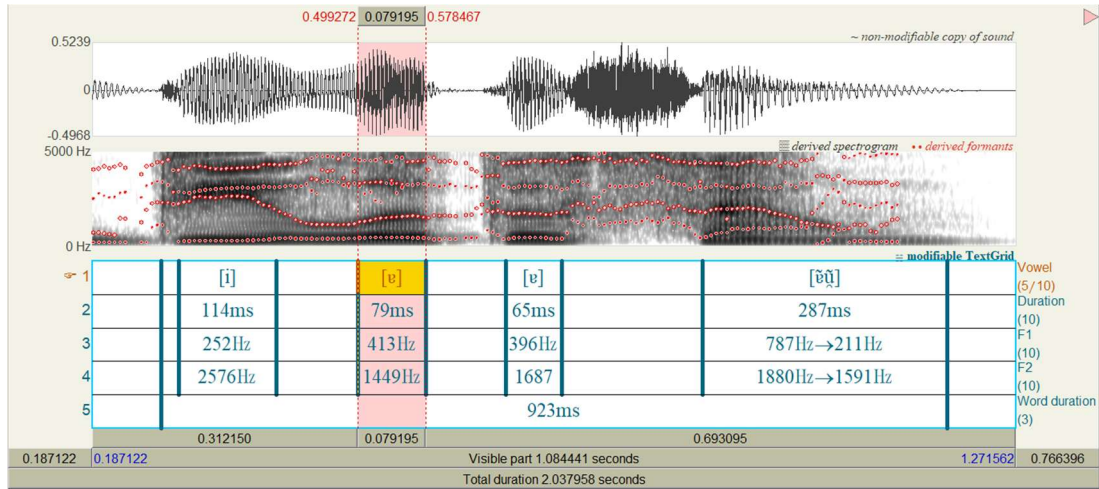
(120) *dilação* [dilaˈsẽũ]



¹³ As of 5 June 2025, the pronunciations of *dilatação* and *dilação* can be accessed for hearing via the *Infopédia Dicionários Porto Editora* at the following entries:

<https://www.infopedia.pt/dicionarios/lingua-portuguesa/dilata%C3%A7%C3%A3o>
<https://www.infopedia.pt/dicionarios/lingua-portuguesa/dila%C3%A7%C3%A3o>

(121) *dilatação* [dile̞tə'sẽũ]



(122)

Syllable	Vowel	Duration (ms)	F1 (Hz)	F2 (Hz)
-la-	[a]	157	846	1653

(123)

Syllable	Vowel	Duration (ms)	F1 (Hz)	F2 (Hz)
-la-	[e]	79	413	1449

The validity of this morpho-phonological interpretation remains open to further investigation, but it seems clear that forms in which **the radical vowel *a* resists reduction tend to lack a thematic vowel**, and that **this absence is closely related to the lack of reduction**.

(iv) Derived words with root-final vowels in hiatus with a preceding vowel

The root-final vowels /^ue/ and /ⁱe/, when placed in hiatus such as -*ae-*, -*ie-*, -*oe-*, -*ue-* – e.g. in *baetilha*, *ajaezar*, *alienar*, *enviesar*, *admoestar*, *amoedar*, *duetista*, and *duelista* – generally do not undergo reduction in contemporary EP, even when they do not bear stress in derivation or verbal inflexion, as shown in (124):

(124) *b*[v'e:]*t-a* ≈ *b*[v.e]*t-ilh-a*; *j*[v'e:]*z* ≈ *a-j*[v.e]*z-a-r*; *al*[i'e:]*n-a* ≈ *al*[i.e]*n-a-r*;

v[i'e:]*s* ≈ *en-v*[i.e]*s-a-r*; *adm*[u'e:]*st-a* ≈ *adm*[u.e]*st-a-r*; *m*[u'e:]*d-a* ≈ *a-m*[u.e]*d-a-r*;

d[u'e:]*t-o* ≈ *d*[u.e]*t-ist-a*, *d*[u.e]*t-ist-ic-o*; *d*[u'e:]*l-o* ≈ *d*[u.e]*l-ist-a*

On the other hand, in Fonseca 1957, many of these examples are transcribed with reduced vowels such as [i] or [ɪ] — e.g. *a-j[ɐ.ɪ]z-ar*, *b[ɐ.ɪ]t-ilha*, *adm[u.ɪ]st-ar*, *a-m[u.ɪ]d-ar*, *d[u.ɪ]l-ita* — suggesting that the phenomenon whereby unstressed vowels remain unreduced in hiatus contexts is likely to be in progress diachronically. Furthermore, the vowel of the inchoative suffixes *-ec-* and *-esc-* consistently shows reduction after lower vowels in all sources — e.g. *esm[ɐ'ɛ:]ce ≈ esm[ɐ.ɪ]cer*, *esv[ɐ'ɛ:]ce ≈ esv[ɐ.ɪ]cer* — but not after more closed vowels, as in *aqu[i'ɛ:]sce ≈ aqu[i.ɛ]scer*. This suggests that it is not hiatus alone that determines whether a vowel undergoes reduction, but also whether it belongs to a root or an affix, as well as the aperture relation between the preceding and following vowels. This is therefore a case that calls for further detailed investigation.

(iv) Metrical foot structure and vowel reduction

The present study, as outlined above, has accounted for vowel reduction or its absence by means of the underlying or derived mora count of vowels, the presence or absence of primary stress in syllables headed by those vowels, and the theoretical apparatus of constraint hierarchies operating within the Vowel Quality Modification Stratum and the Vowel Quantity Adjustment Stratum, respectively. There still remain, however, phenomena that cannot be explained by these factors alone. For instance, *ecónom-o* "manager; steward", *económ-ico* "economic", and *econom-ia* "economy", which all share the word constituent *econom-*, exhibit the following vowel alternations, as shown in (125):

$$(125) \text{ } ec[\acute{\text{ɔ}}:]n[u]m-o \approx ec[u]n[\acute{\text{ɔ}}:]m-ico \approx ec[\text{ɔ}]n[u]m-ia$$

Judging from the alternation $ec[\acute{\text{ɔ}}:]n[u]m-o \approx ec[u]n[\acute{\text{ɔ}}:]m-ico$, the underlying form of the radical in these words appears to be $/ik\acute{\text{ɔ}}n\acute{\text{ɔ}}m-/$, where the first $/\acute{\text{ɔ}}/$ is, like the second $/\acute{\text{ɔ}}/$, an underlying short vowel. In $ec[\text{ɔ}]n[u]m-ia$, nevertheless, this underlying short vowel $/\acute{\text{ɔ}}/$ is realised as $[\text{ɔ}]$, showing no reduction. One possible explanation lies in the difference in prosodic structure, as illustrated in (126):

$$(126) (,i.k\acute{\text{ɔ}})('\text{n}\acute{\text{ɔ}}.mi)ko \approx i(,k\acute{\text{ɔ}}.\text{n}\acute{\text{ɔ}})('\text{mi}.a)$$

In *económico*, where the underlying $/\acute{\text{ɔ}}/$ is realised as reduced $[u]$, the vowel occurs in the complement syllable in the foot $(,i.k\acute{\text{ɔ}})$. In contrast, in *economia*, where the same underlying $/\acute{\text{ɔ}}/$ is realised as unreduced $[\text{ɔ}]$, it occurs in the head syllable of the foot $(,k\acute{\text{ɔ}}.\text{n}\acute{\text{ɔ}})$. This prosodic difference appears to be related to whether vowel reduction takes place, suggesting that further

investigation and research on this point is required.

7. Conclusion

Although the concept of mora has not traditionally been applied to the analysis of Portuguese vowels, the present study adopts this notion and posits an opposition between the underlying short vowels $/i^{\mu} e^{\mu} \varepsilon^{\mu} a^{\mu} \circ^{\mu} o^{\mu} u^{\mu}/$ and the underlying long vowels $/\varepsilon^{\mu\mu} a^{\mu\mu} \circ^{\mu\mu}/$ in Contemporary European Portuguese. Furthermore, by establishing a system of vowel quality reduction and syllable/vowel quantity adjustment arising from the constraint hierarchies operating at the Vocalic Quality Modification Stratum and the Vocalic Quantity Adjustment Stratum, we attempted to account for two types of vowel alternation — $[i: \acute{e}: \acute{e}: \acute{a}: \acute{o}: \acute{o}: \acute{u}:] \approx [i \text{ } \text{ɨ} \text{ } \text{ɨ} \text{ } u \text{ } u \text{ } u]$ and $[\acute{e}: \acute{a}: \acute{o}:] \approx [\varepsilon \text{ } a \text{ } \circ]$ — observed at the phonetic level. Building on this descriptive account, we exhibited that it is also possible to provide a unified explanation for a range of apparently unrelated phonetic phenomena: the absence of vowel quality reduction in nasal vowels in unstressed syllables; the characteristic vowel alternations in open syllables preceding prepalatal and palatal consonants $[\text{ʃ} \text{ } \text{ʒ} \text{ } \text{ʎ} \text{ } \text{n}]$; the phonetic realisation of vowel coalescence at the level of prosodic phonology; the opposition $[\acute{e}:] \approx [\acute{a}:]$ in the verb endings *-amos* and *-ámos* in the first-person plural of the present and preterite indicatives; vowel lowering in radical vowels found in derived nouns and adjectives formed by suffixation to verbal radicals lacking intervening thematic vowels; and the absence of vowel reduction in unstressed syllables of derived words with negative connotations.

Finally, in traditional accounts — whether explicitly or implicitly — it has been assumed that whether an unstressed vowel in a given word undergoes reduction or not is synchronically unpredictable, and must therefore be memorised individually at the lexical level. In other words, speakers are assumed to store not only the "underlying segmental sequence" of each word, but also an additional piece of information regarding whether vowel reduction applies or not. This results in a high degree of lexical redundancy. By contrast, under the present analysis, it suffices in most cases for the speaker to memorise only the underlying segmental sequence of a word at the lexical level. The presence or absence of vowel reduction becomes predictable from the evaluation of output candidates by the constraint rankings operating in each of the strata of phonological derivation. As a result, in most cases, the only information that must be stored is the phoneme sequence along with vowel quantity. This yields a representation that is more compact and principled than in traditional theories, which rely on rote memorisation of

exceptional reduction behaviour. At the same time, however, the existence of alternation patterns such as $ec[\acute{o}]n[u]m-o \approx ec[u]n[\acute{o}]m-ico \approx ec[\omicron]n[u]m-ia$ highlights the need for further investigation and research, as well as for refinement of the theoretical framework.

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